

CSE472: Machine Learning Sessional

Date: 3 January, 2024

Submitted by:
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CSE, L-4/T-2, B2

Models (training and validation)

Model 1(Best):

```
L1=Dense(28*28, 1024)
L2=ReLU()
L3=Dropout(0.25)
L4=Dense(1024,26)

network = [
    L1,
    L2,
    L3,
    L4
]

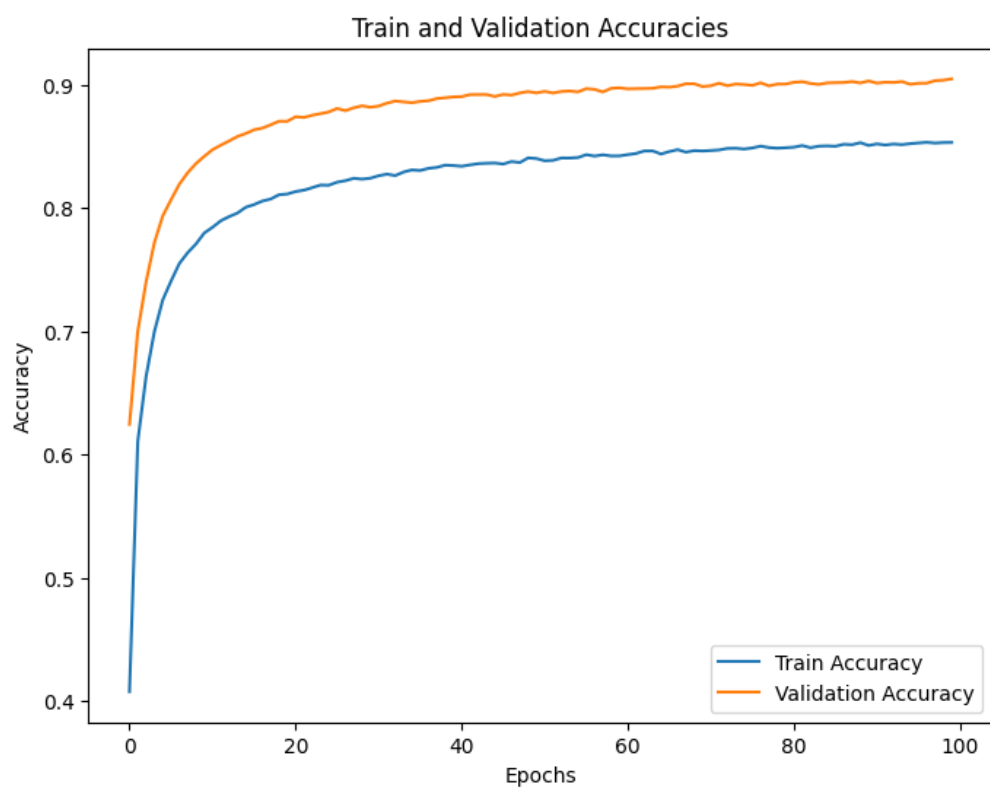
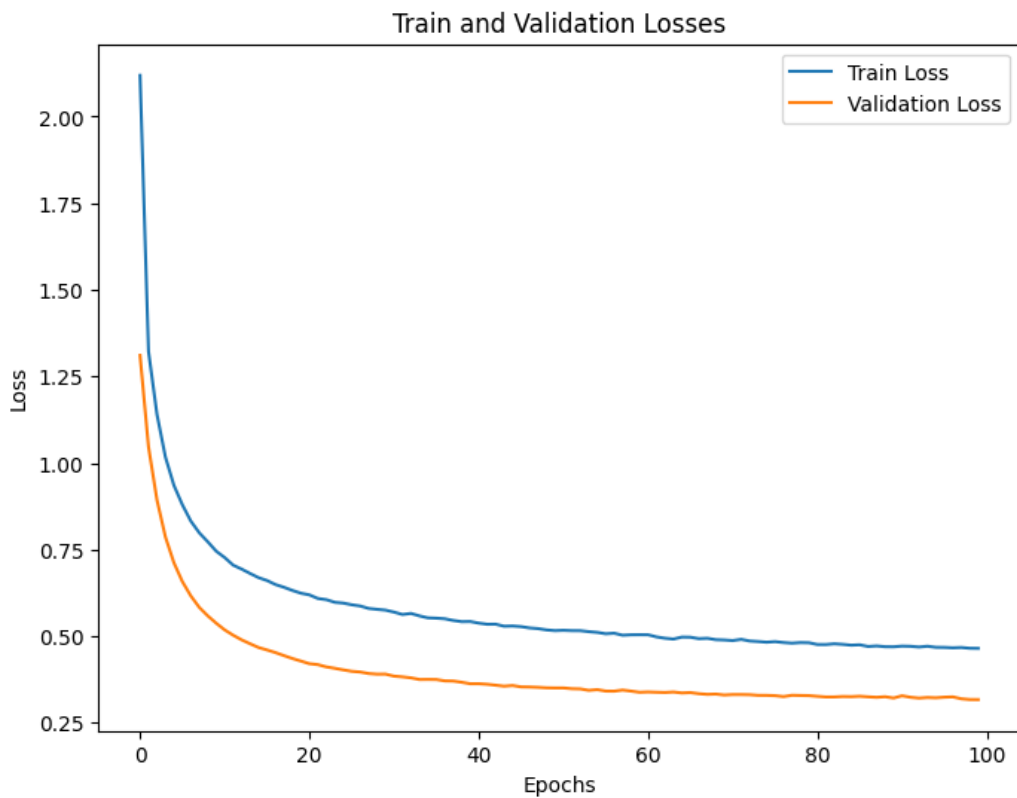
train(network, softmax_crossEntropy(), train_loader, val_loader, epochs=100, learning_rate=0.01)
```

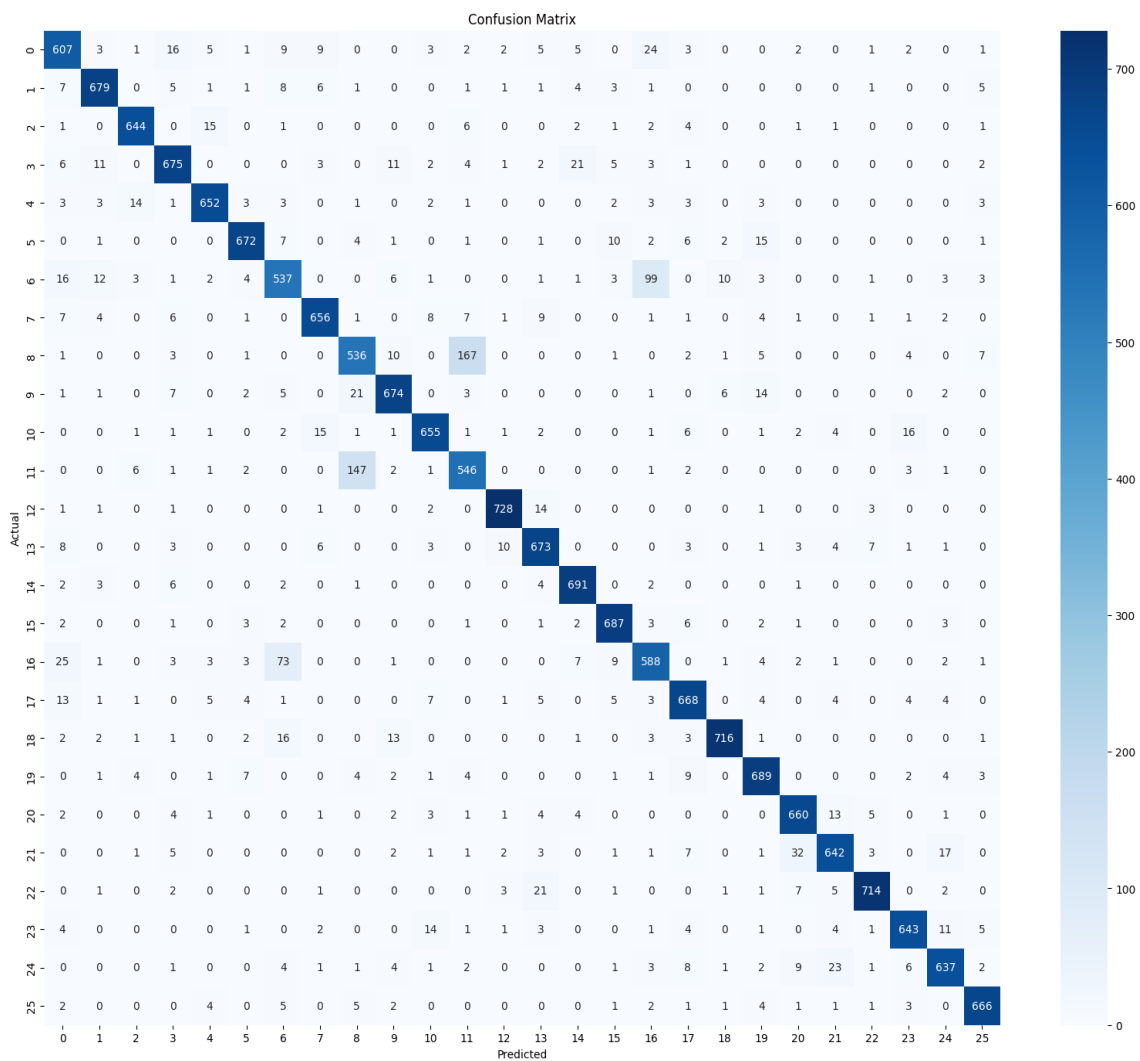
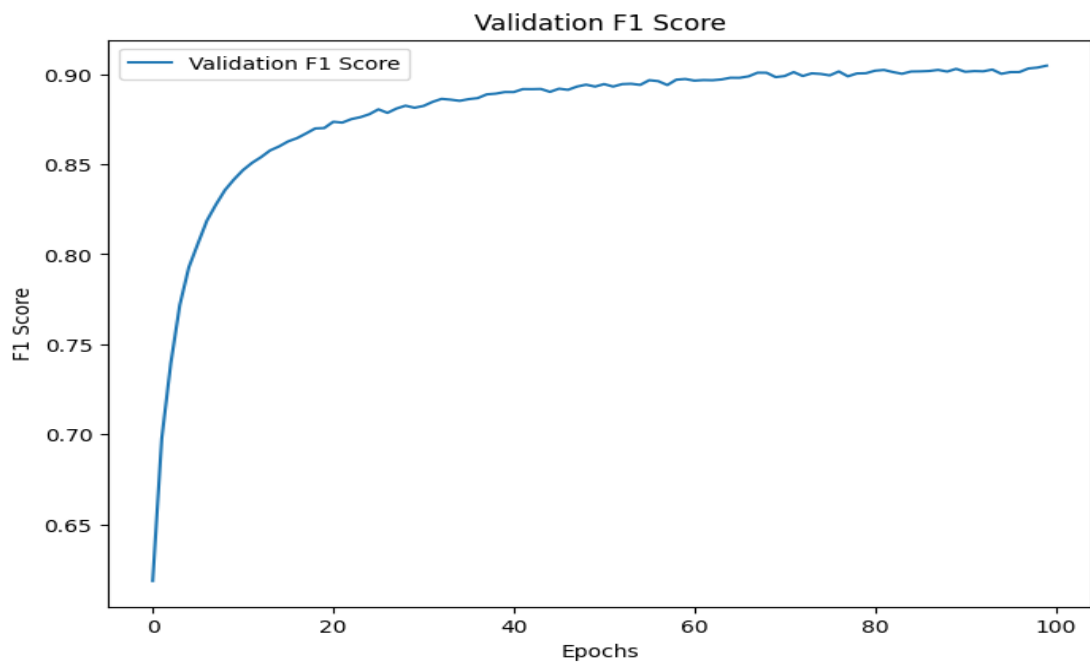
Metrics:

learning rate	loss		accuracy		macro_f1
	training	validation	training	validation	validation
0.01	0.465	0.317	0.853	0.905	0.905
0.005	0.466	0.352	0.856	0.894	0.895
0.001	0.628	0.492	0.814	0.855	0.855
0.0005	0.789	0.641	0.771	0.814	0.814

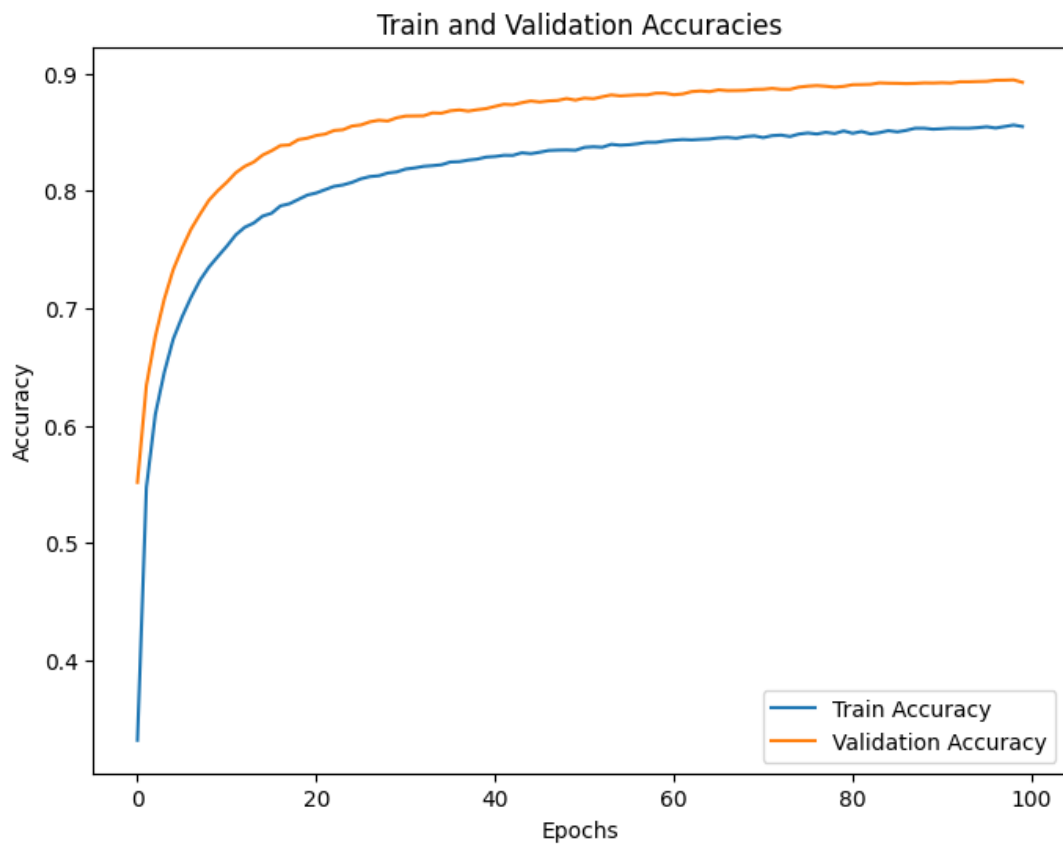
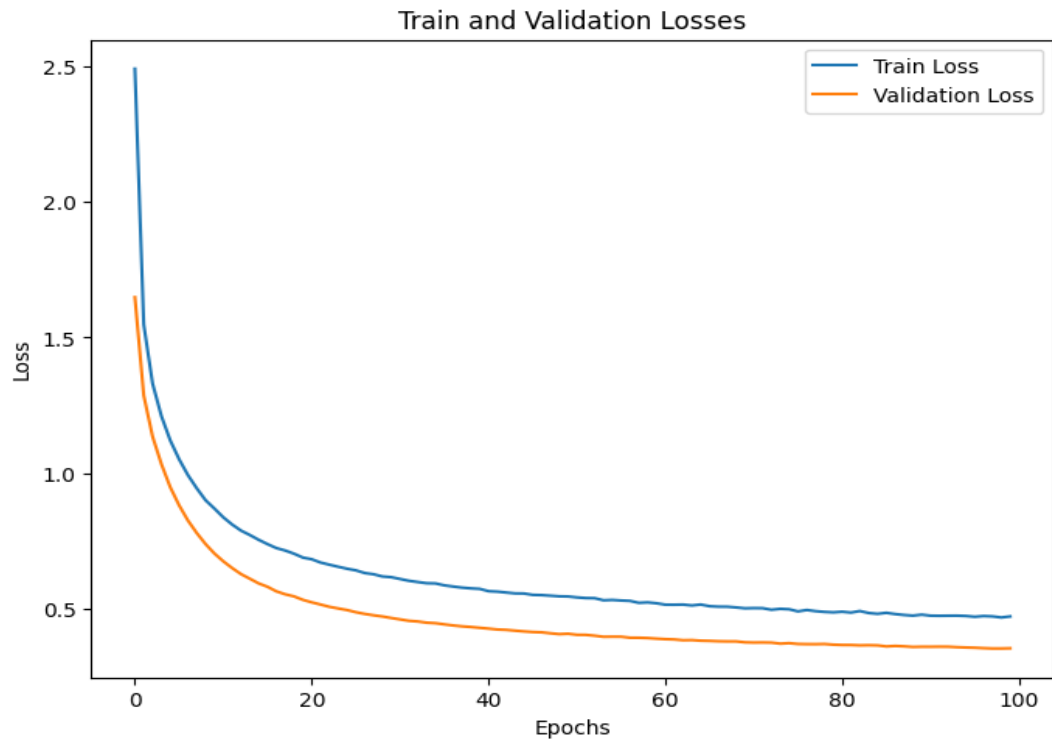
Graphs of Metrics:

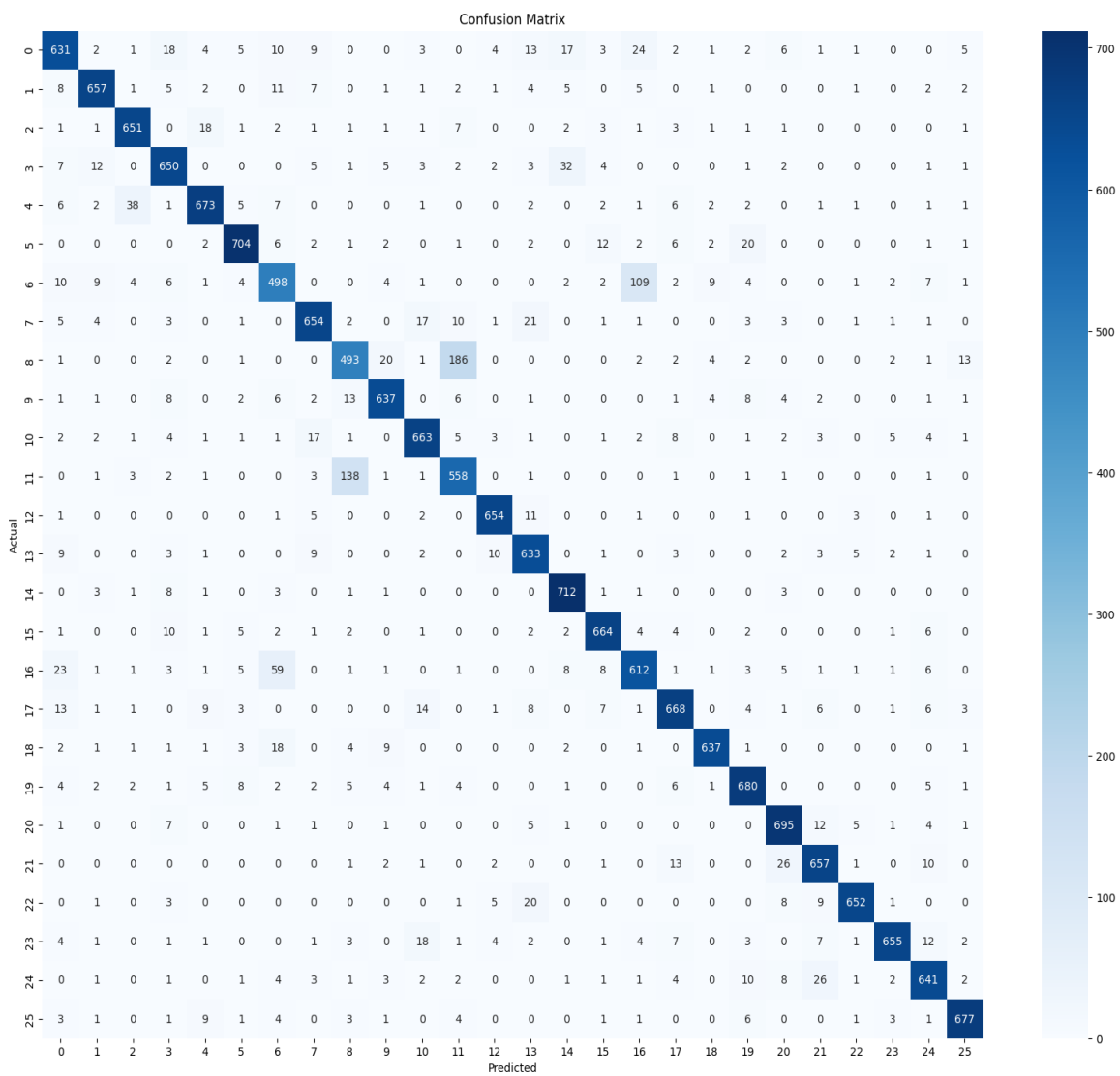
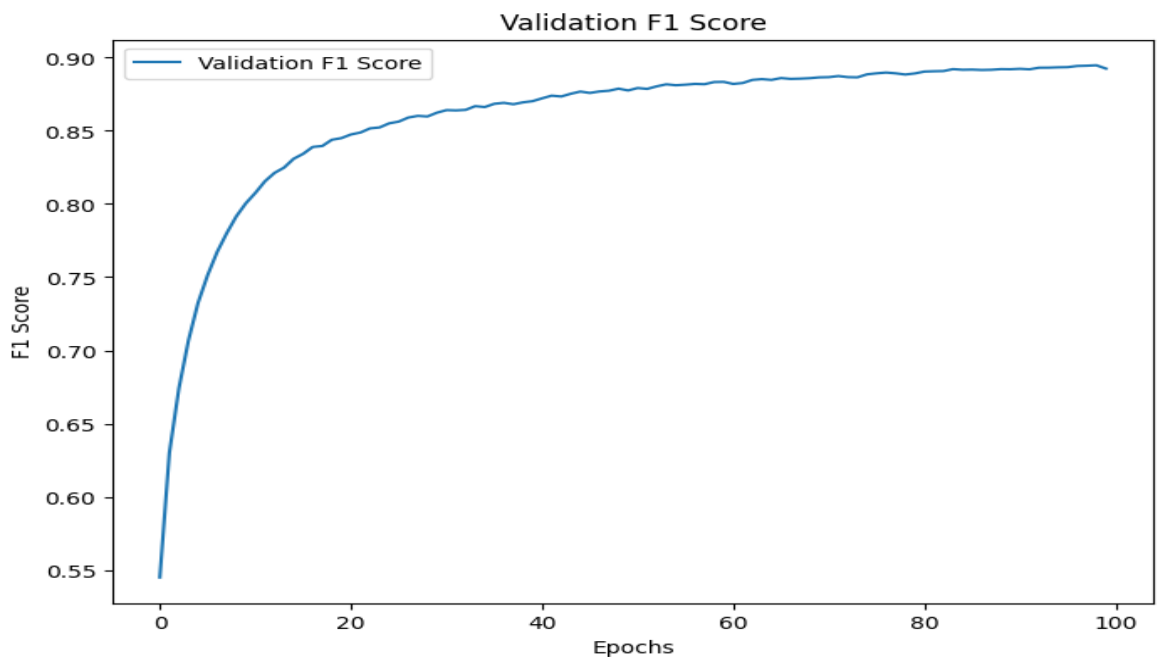
LR = 0.01 (Best)



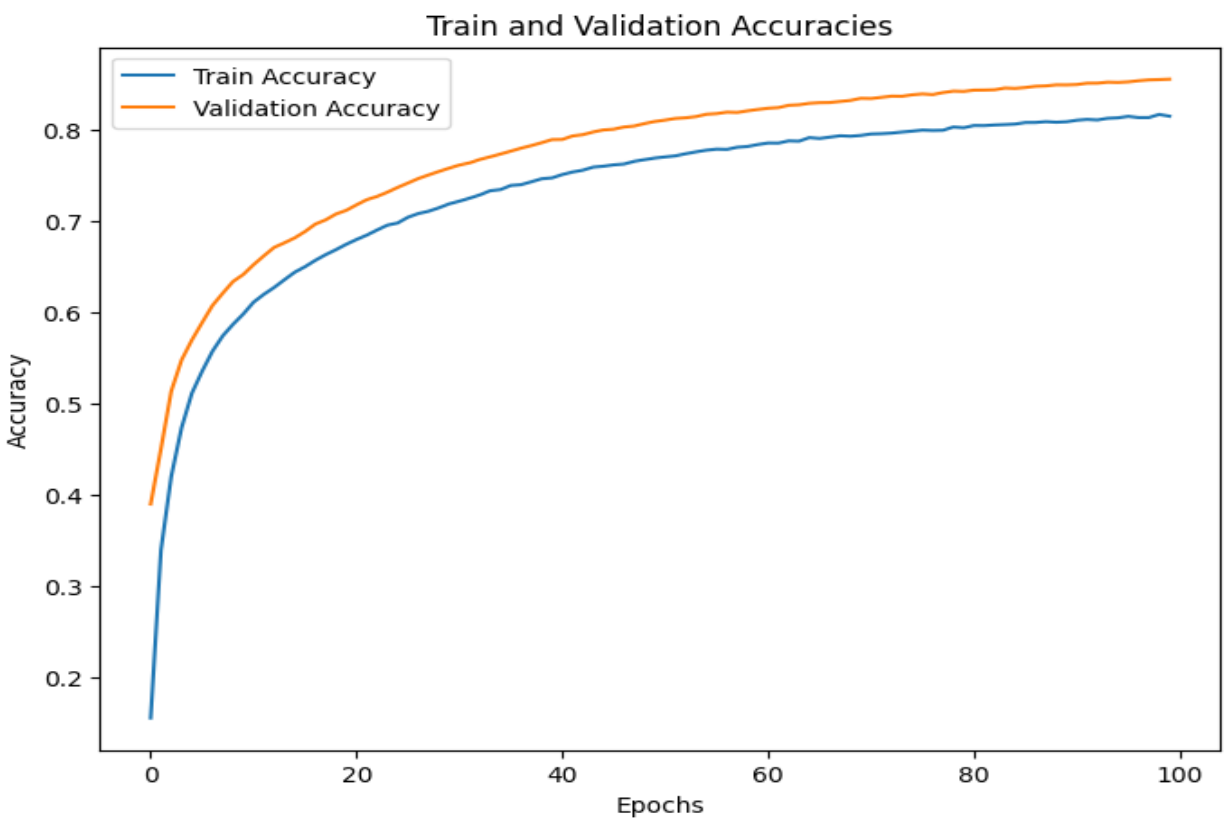
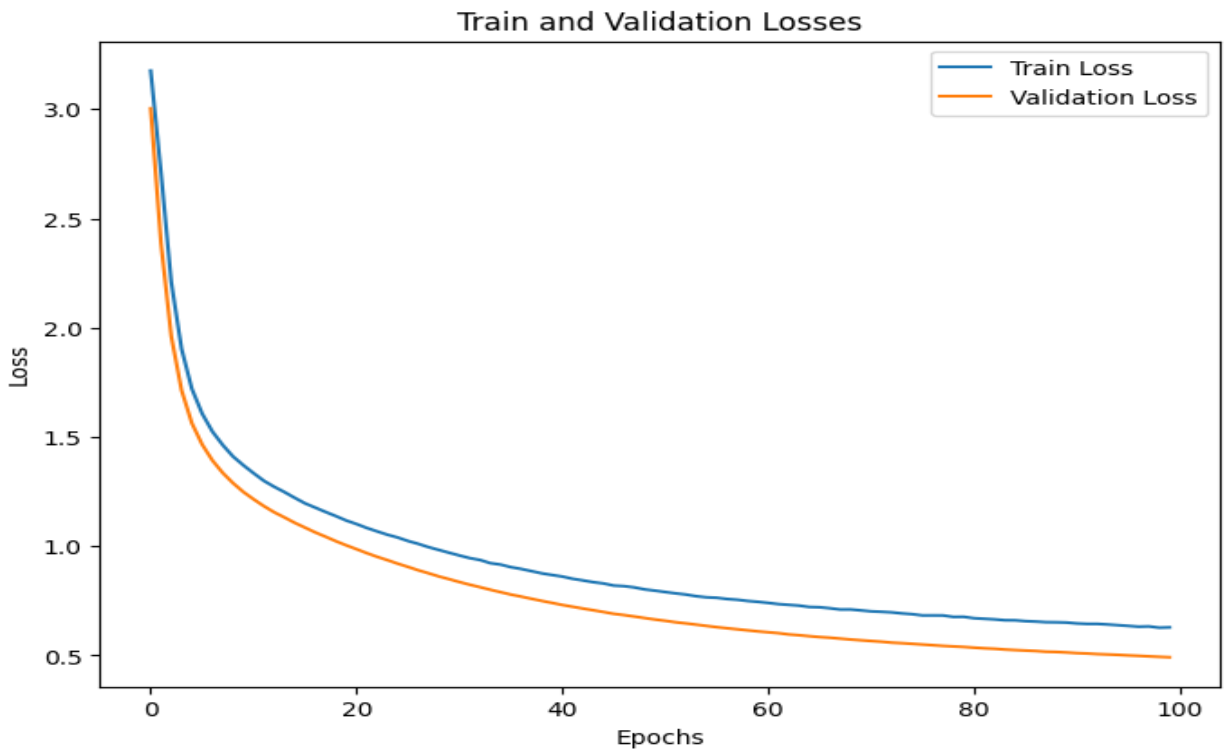


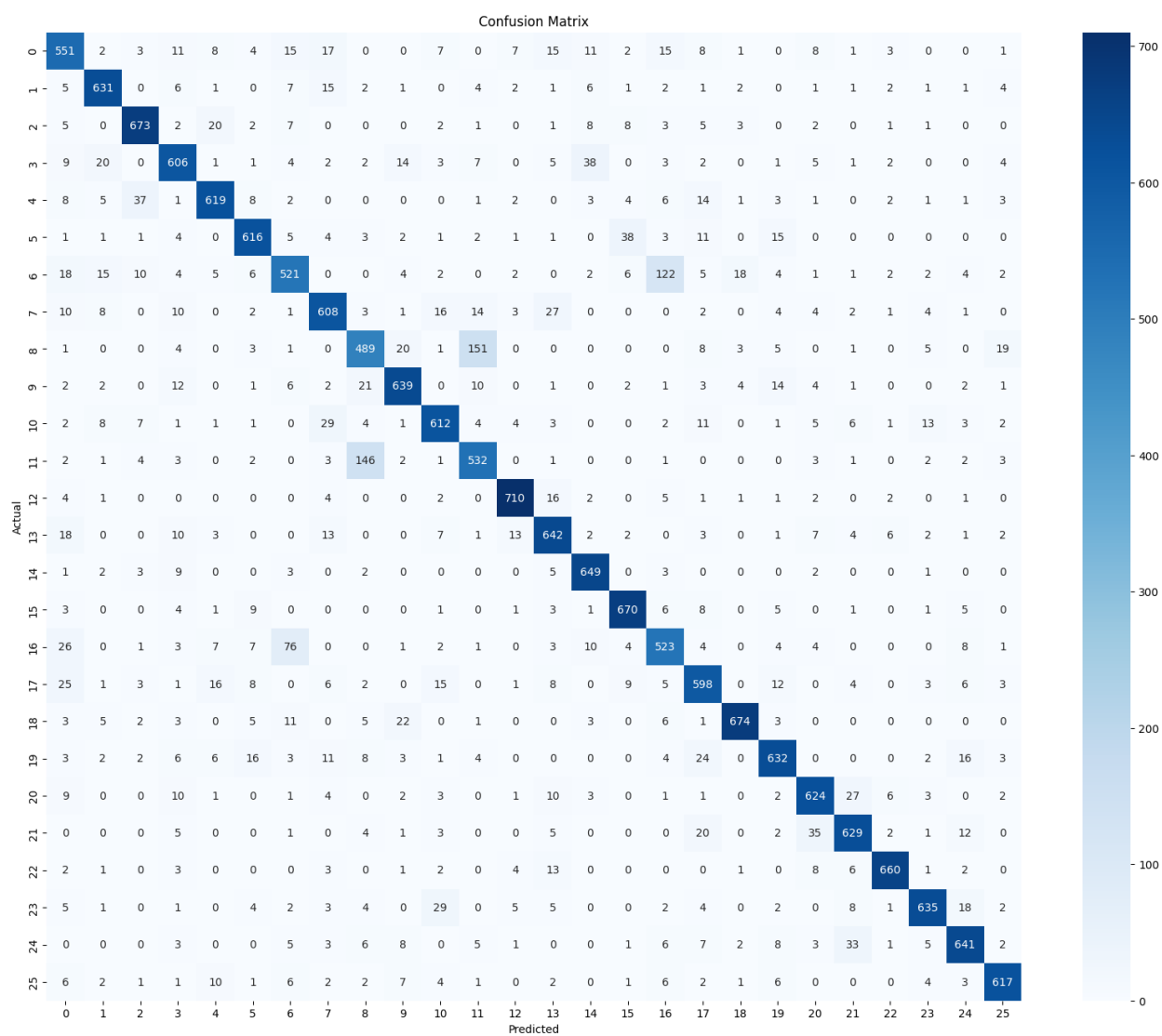
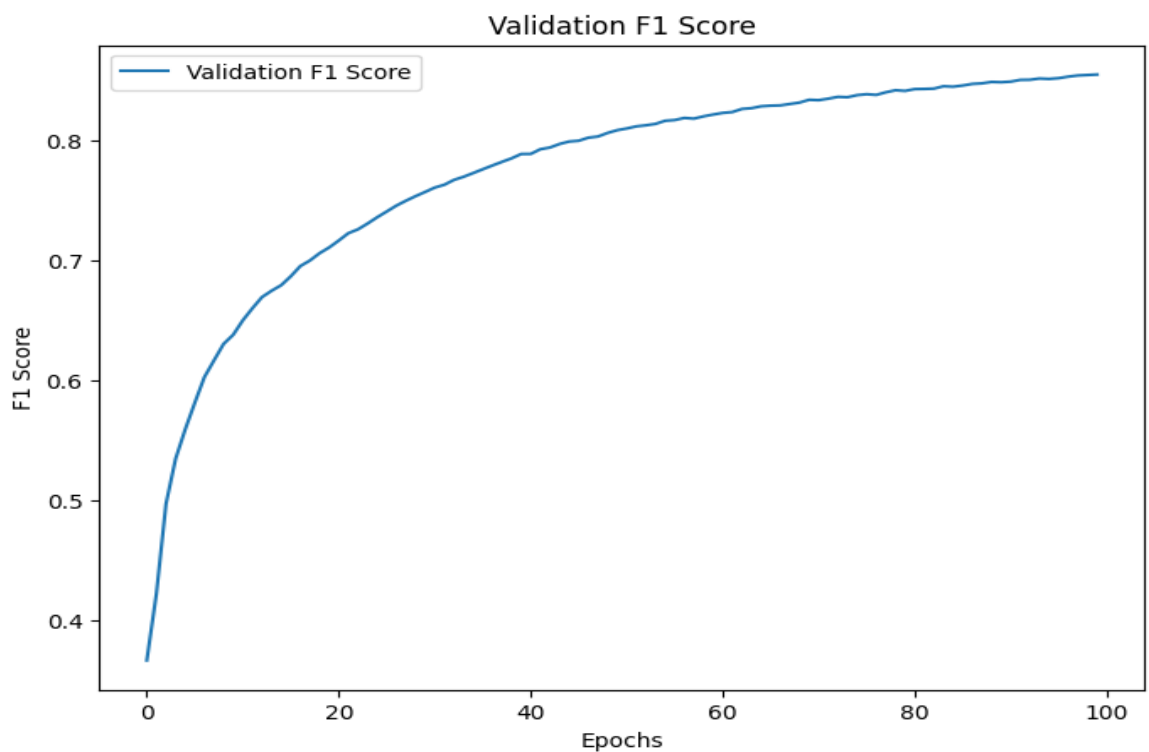
LR = 0.005



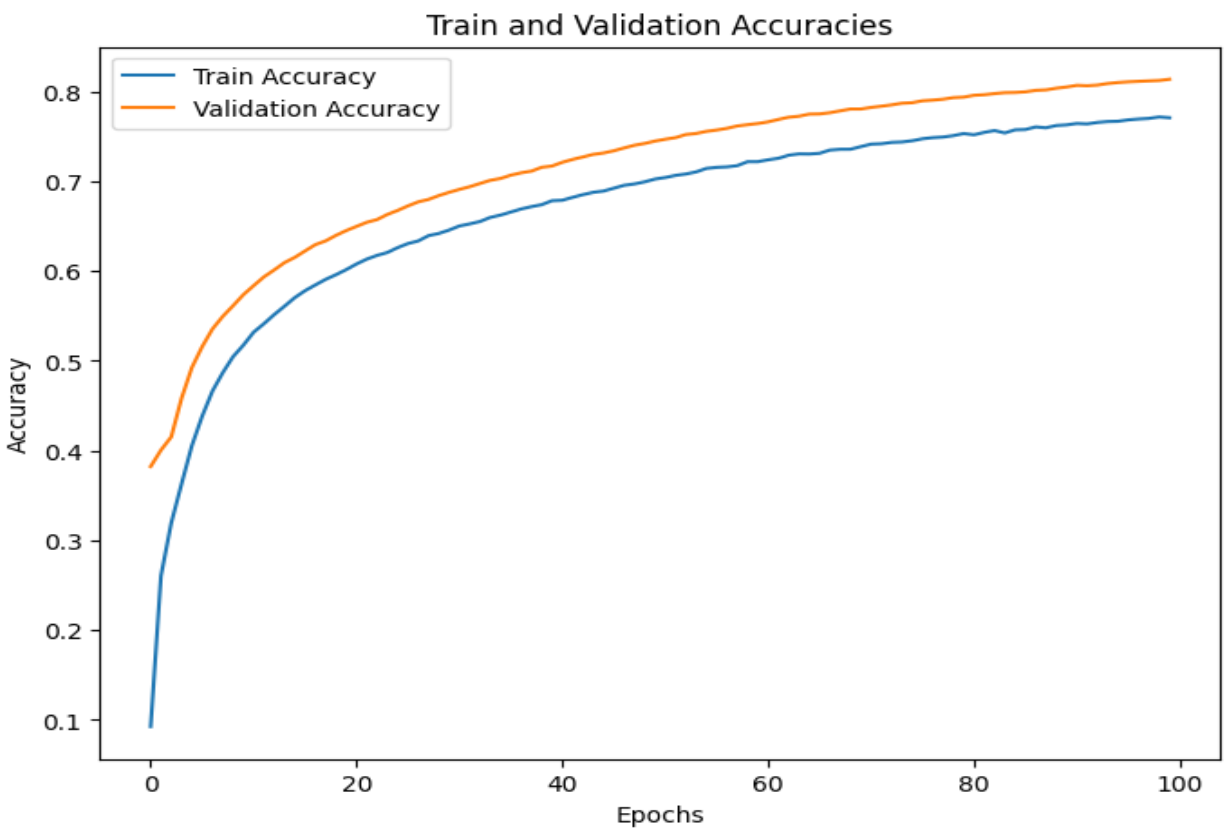
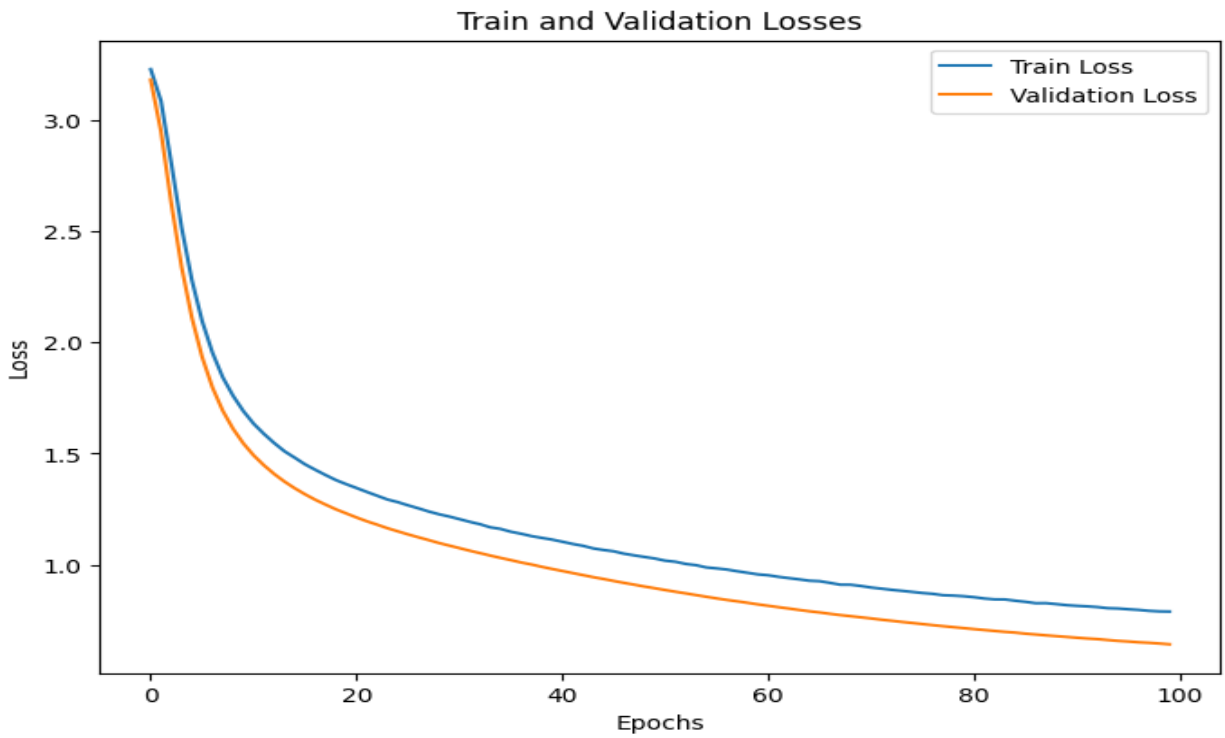


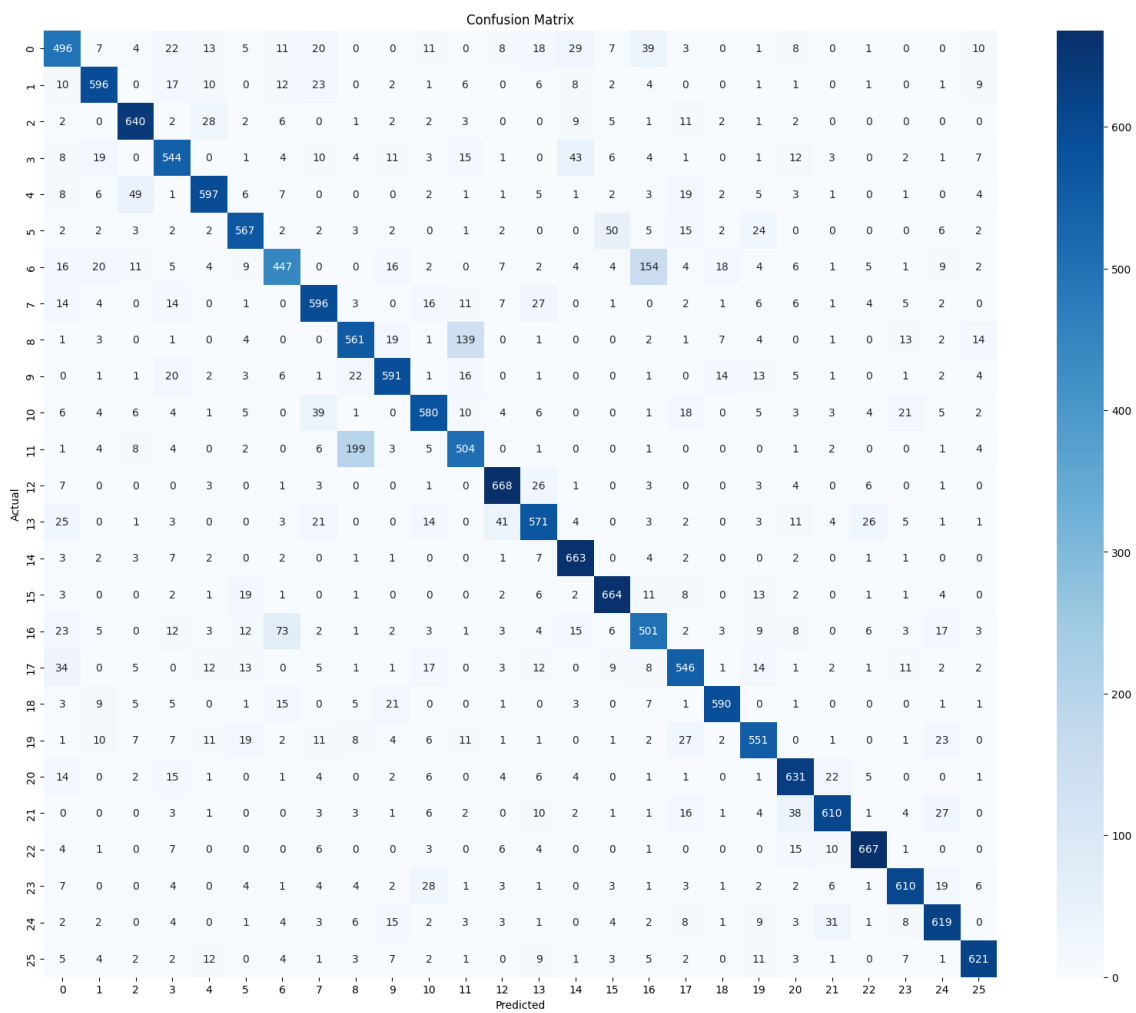
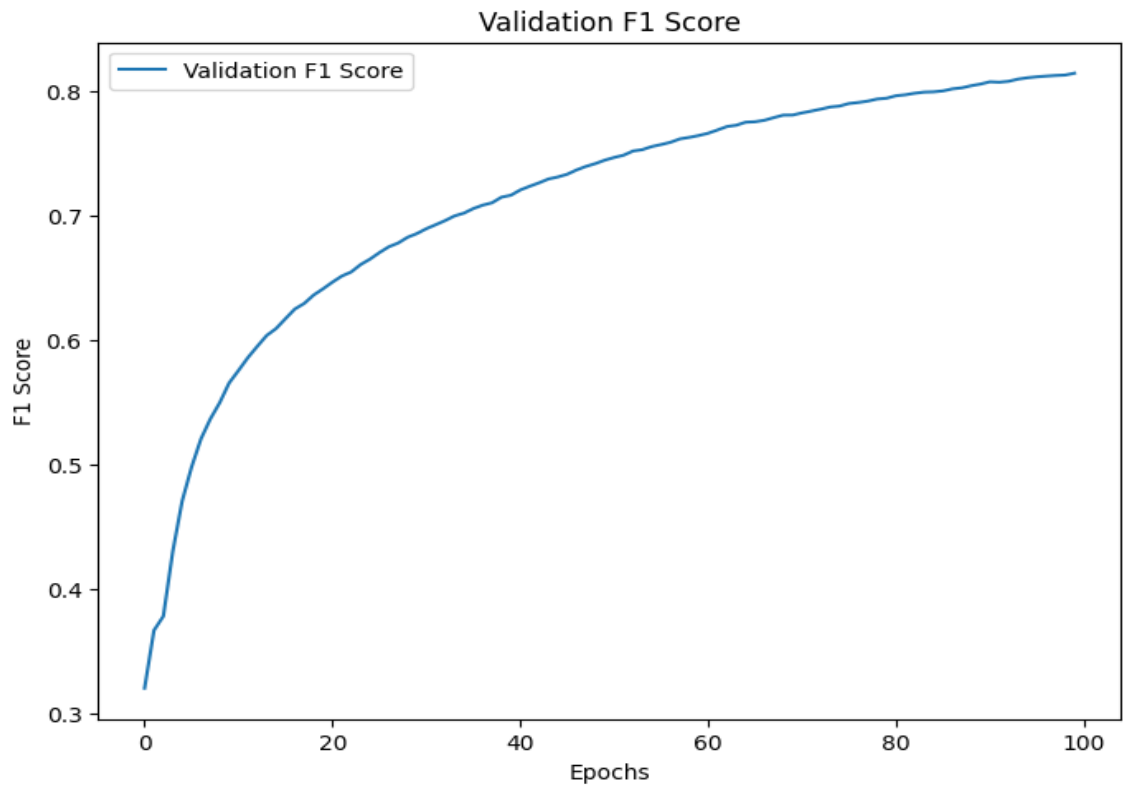
LR = 0.001





LR = 0.0005





Model 2:

```
L1=Dense(28*28, 512)
L2=ReLU()
L3=Dropout(0.25)
L4=Dense(512,26)

network = [
    L1,
    L2,
    L3,
    L4
]

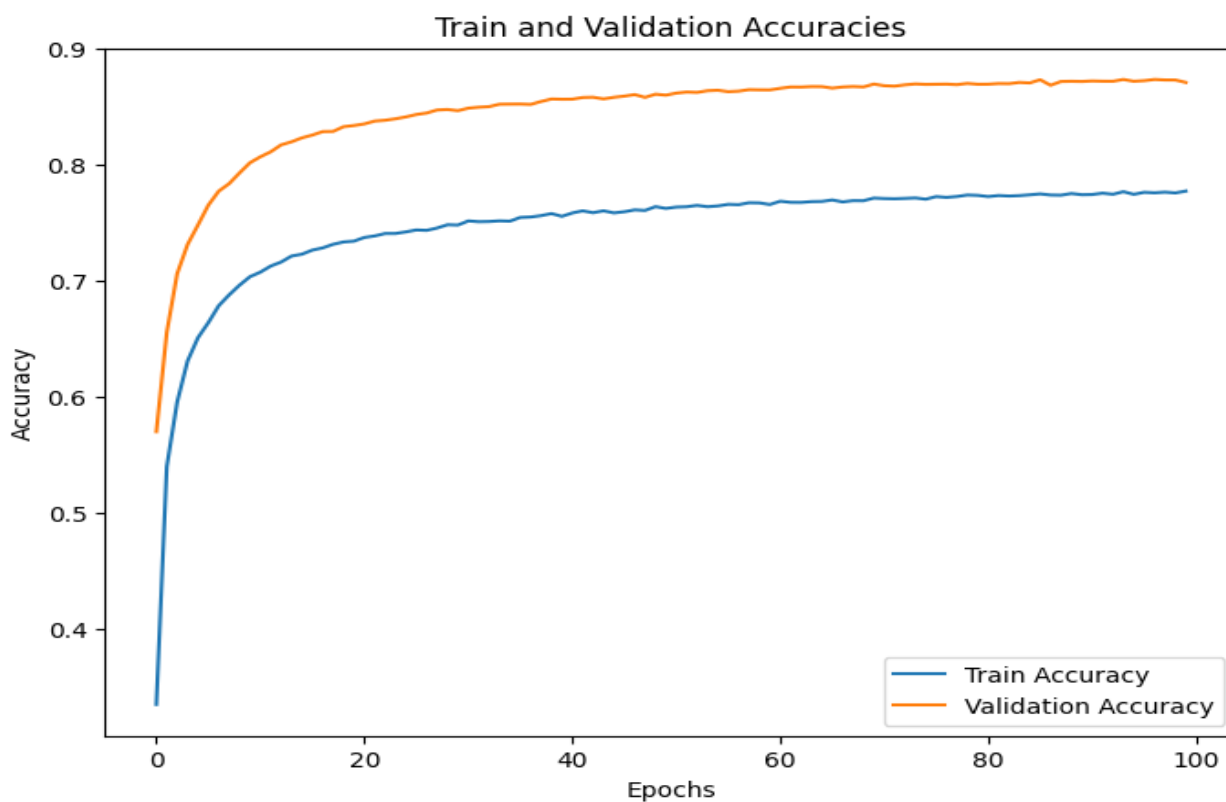
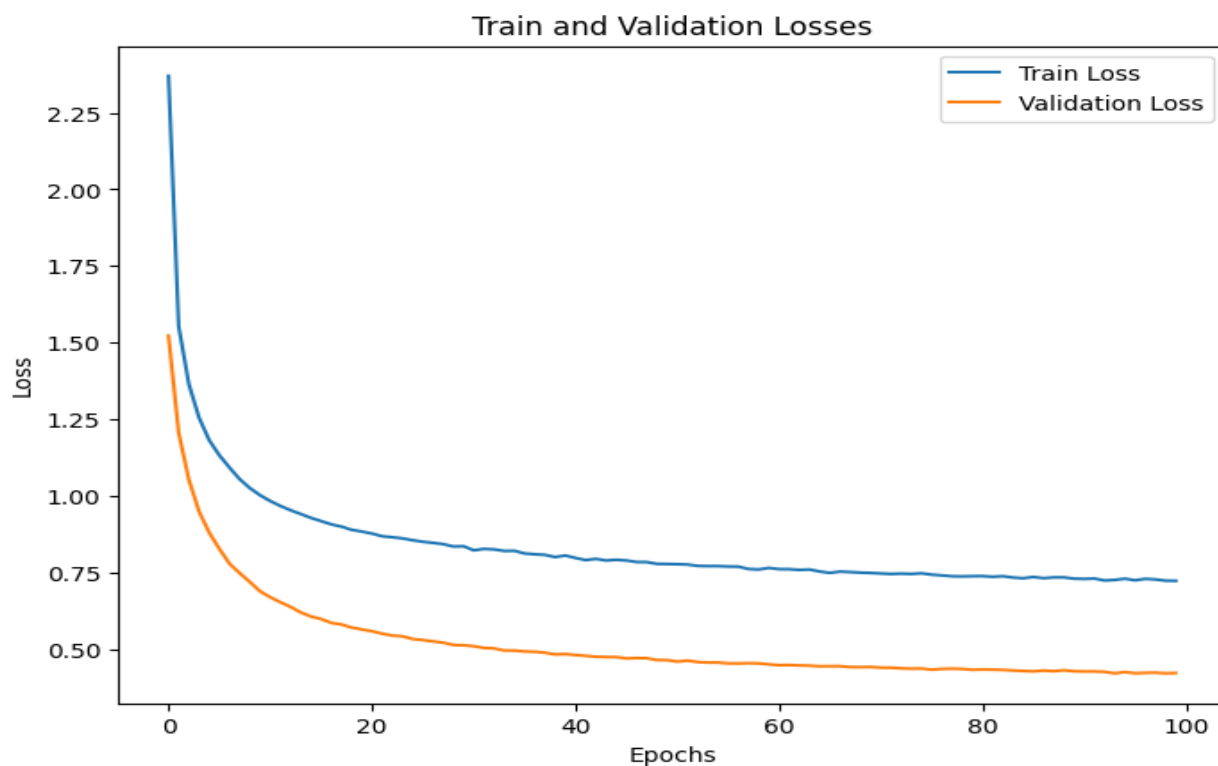
train(network, softmax_crossEntropy(), train_loader, val_loader, epochs=100, learning_rate=0.01)
```

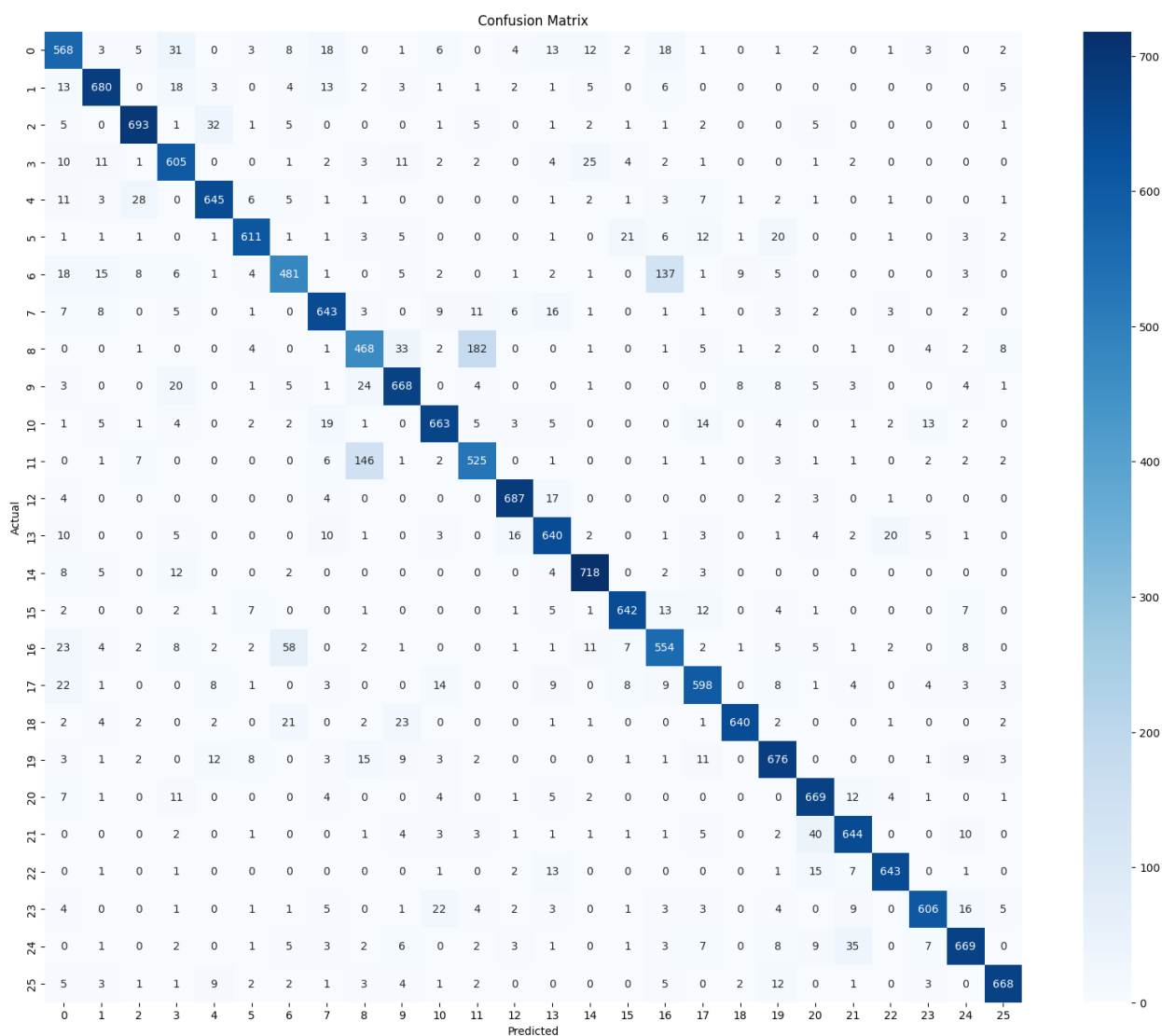
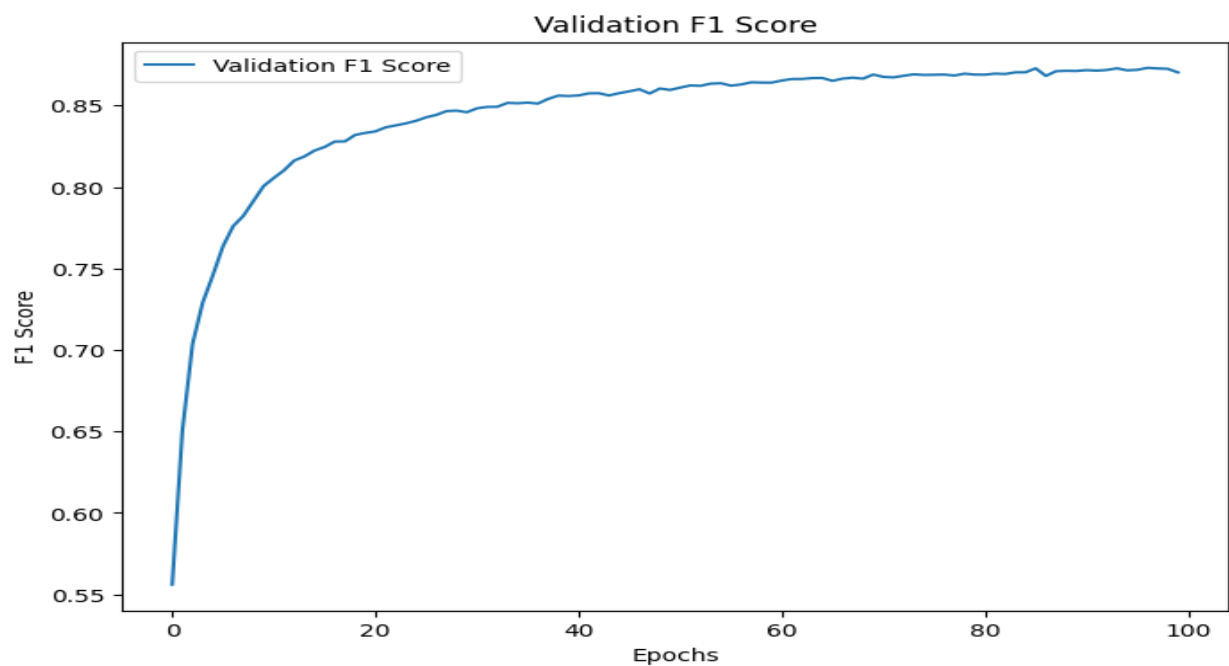
Metrics:

learning rate	loss		accuracy		macro_f1
	training	validation	training	validation	validation
0.01	0.729	0.422	0.776	0.874	0.873
0.005	0.744	0.451	0.774	0.866	0.865
0.001	0.847	0.599	0.751	0.829	0.828
0.0005	1.035	0.788	0.698	0.777	0.776

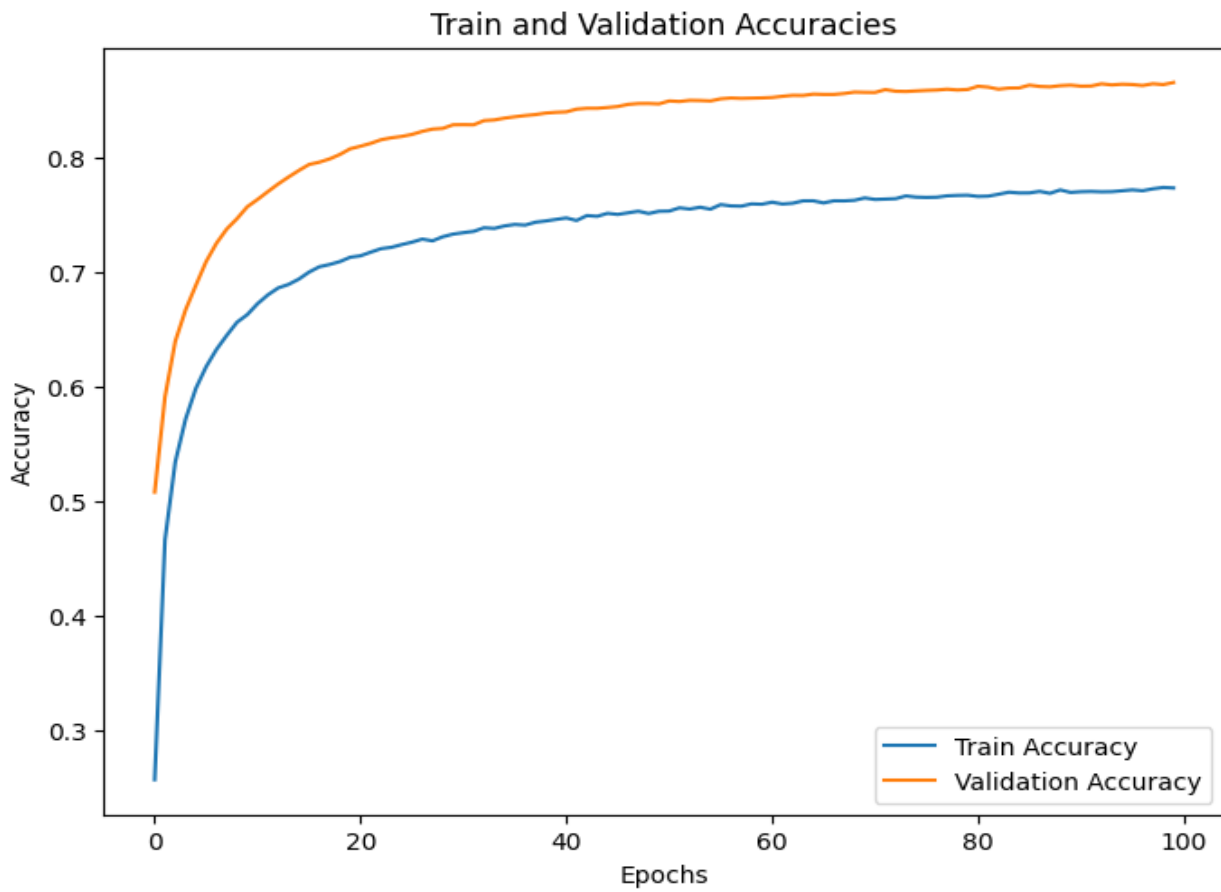
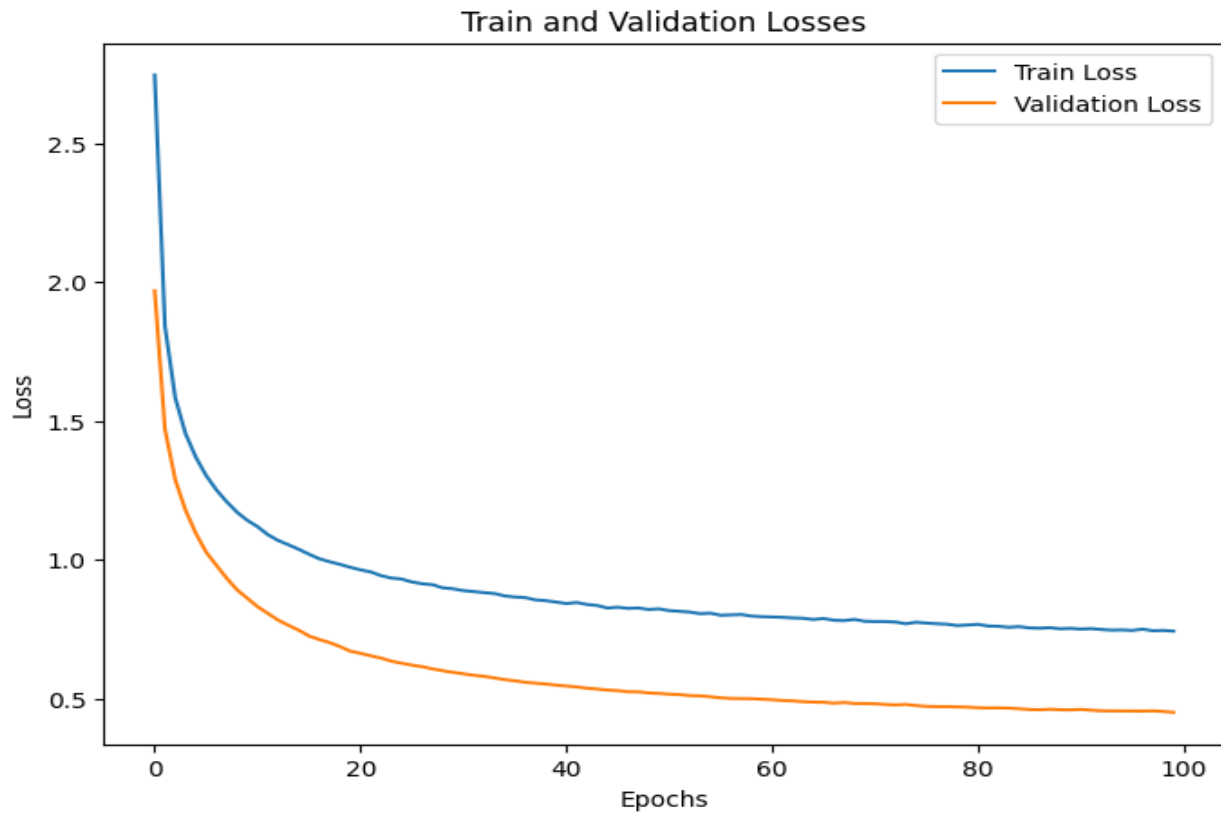
Graphs of Metrics:

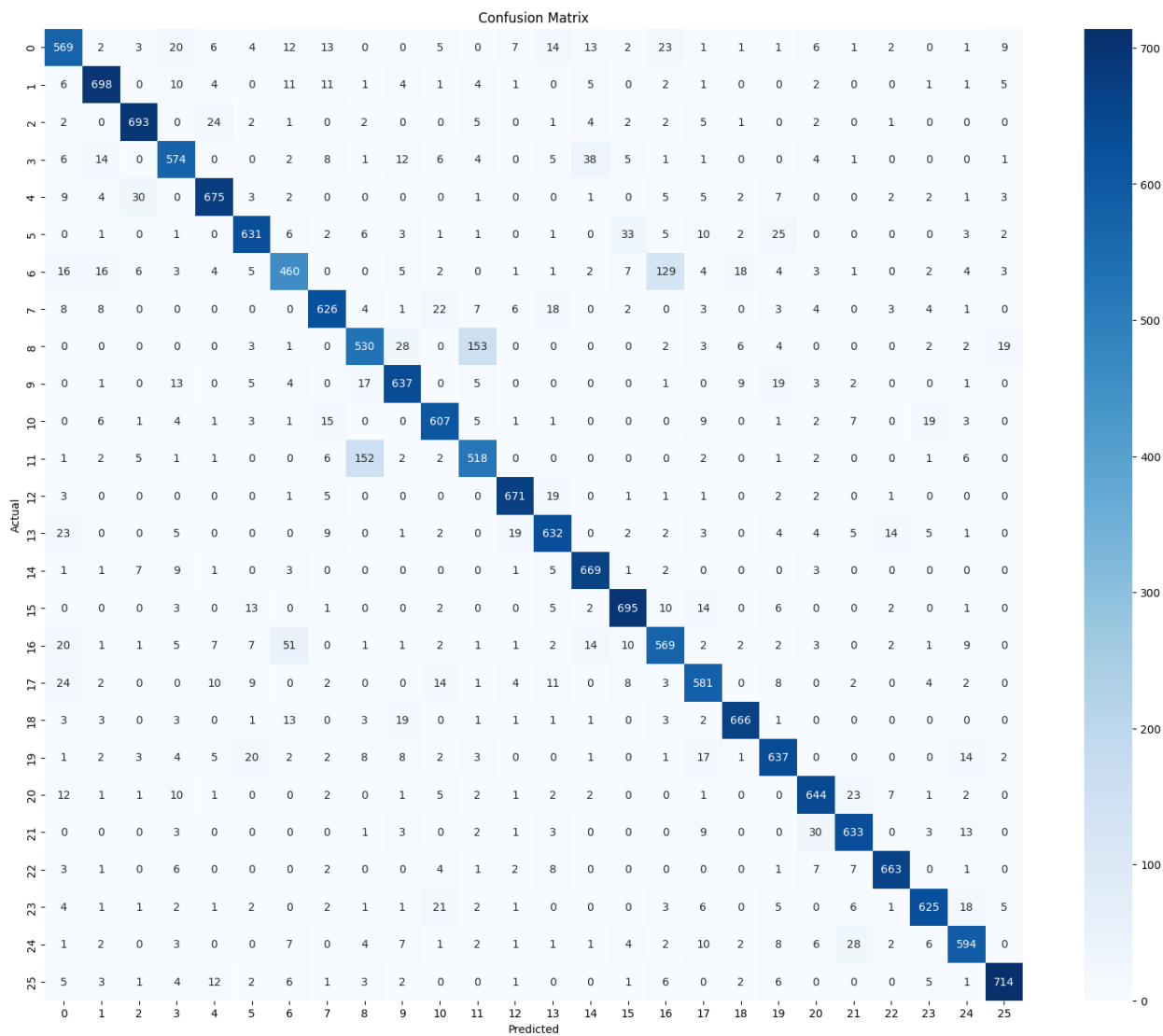
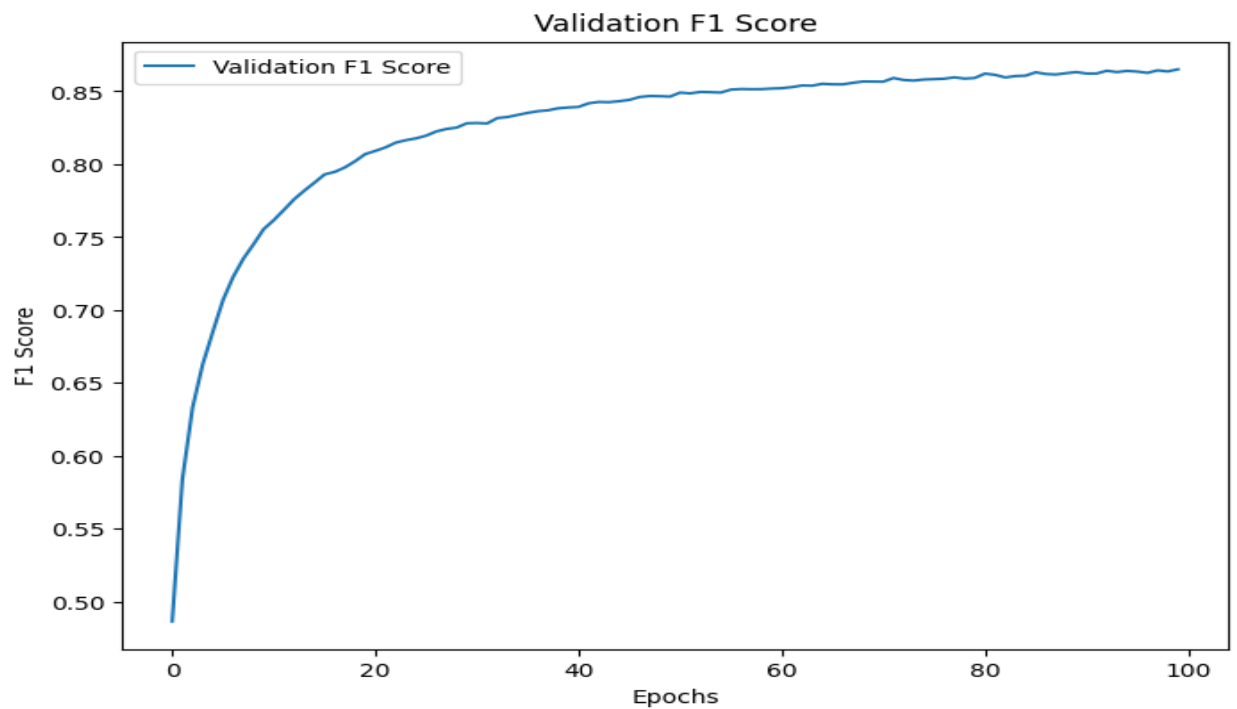
LR = 0.01



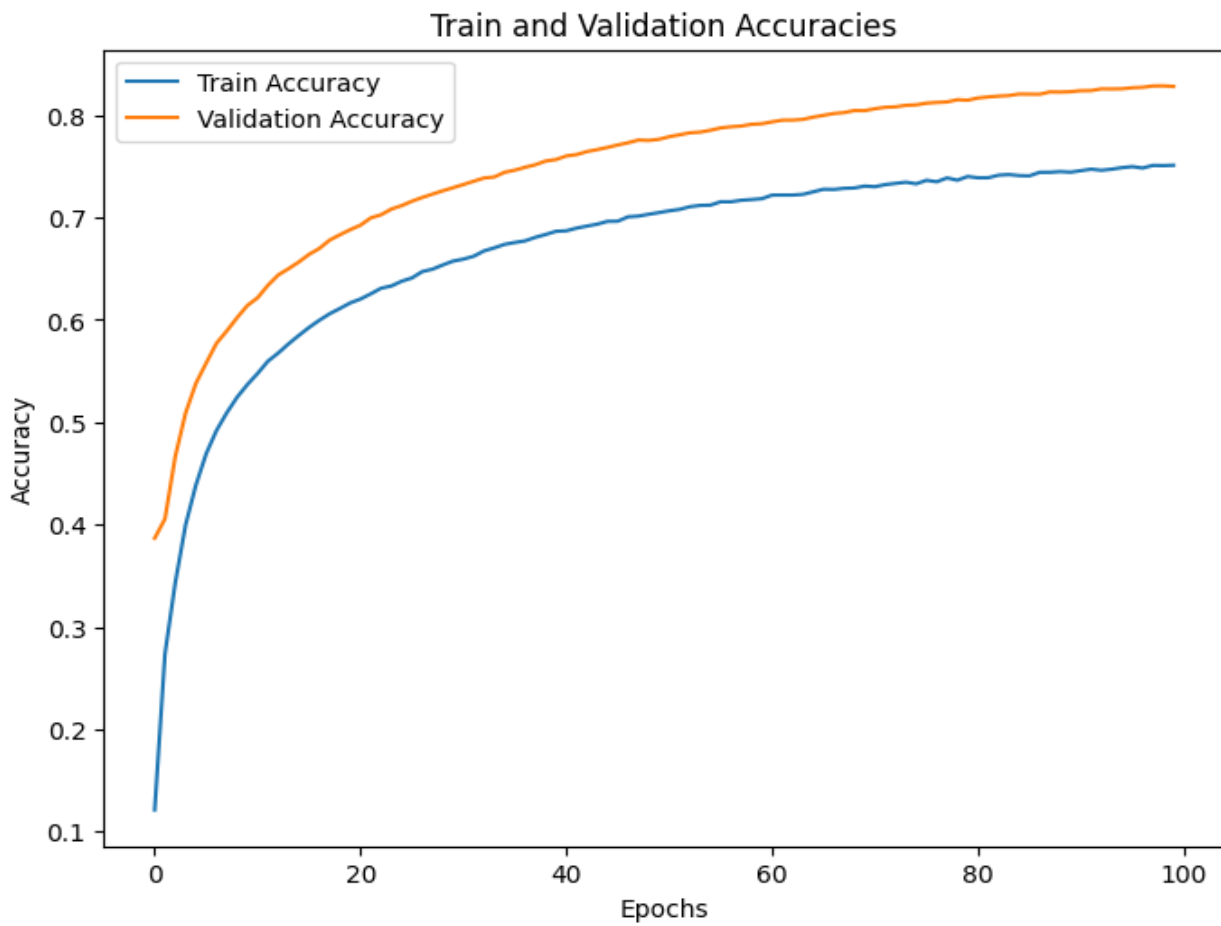
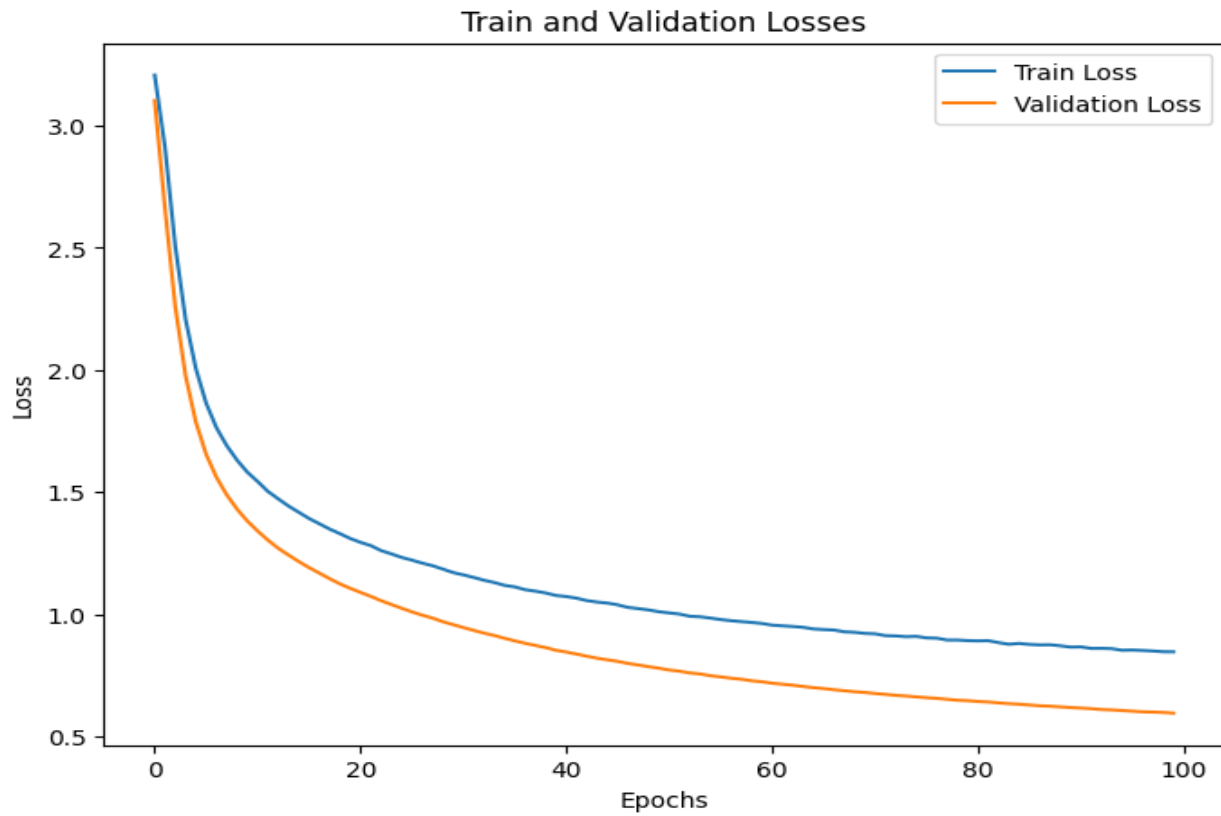


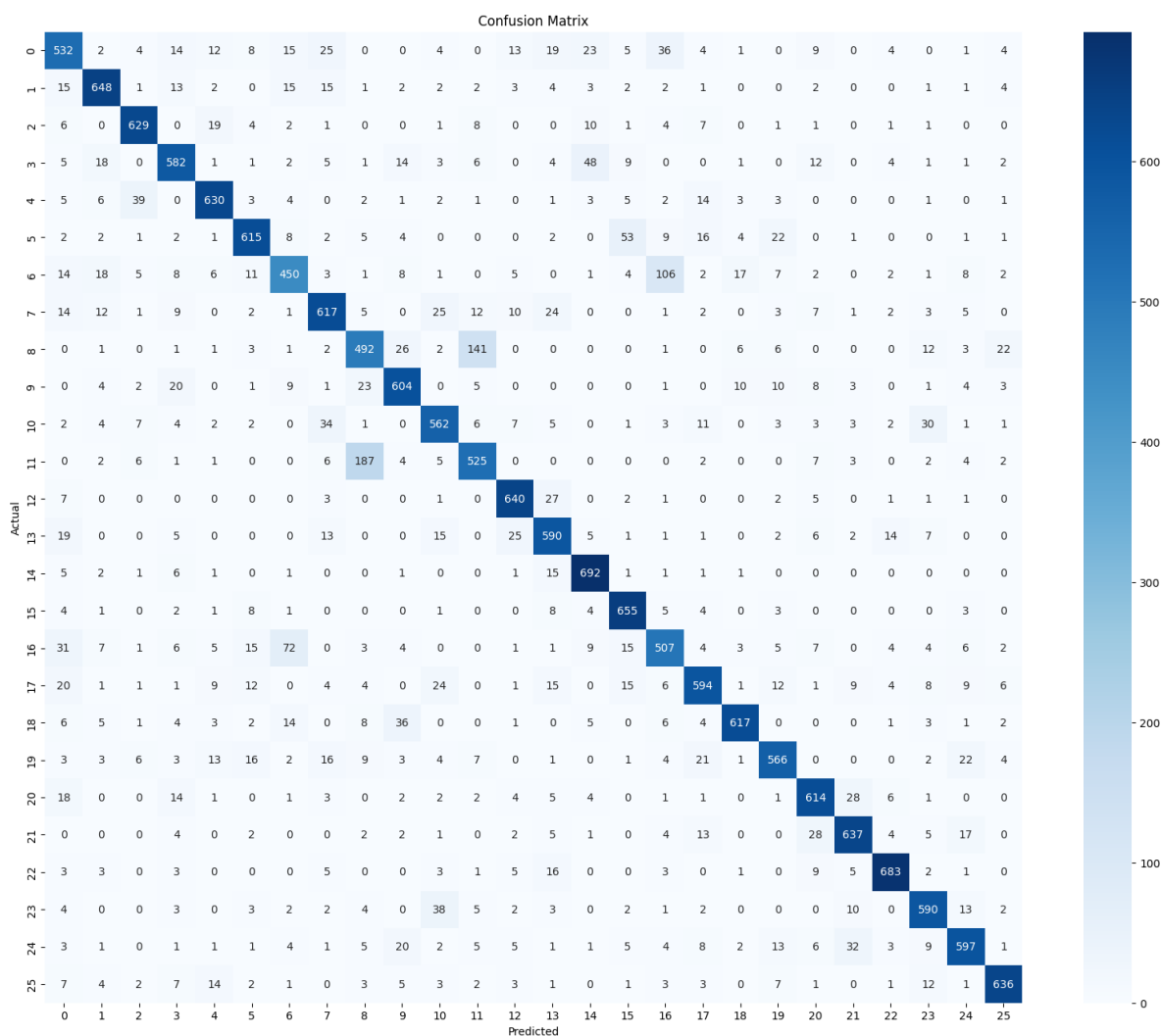
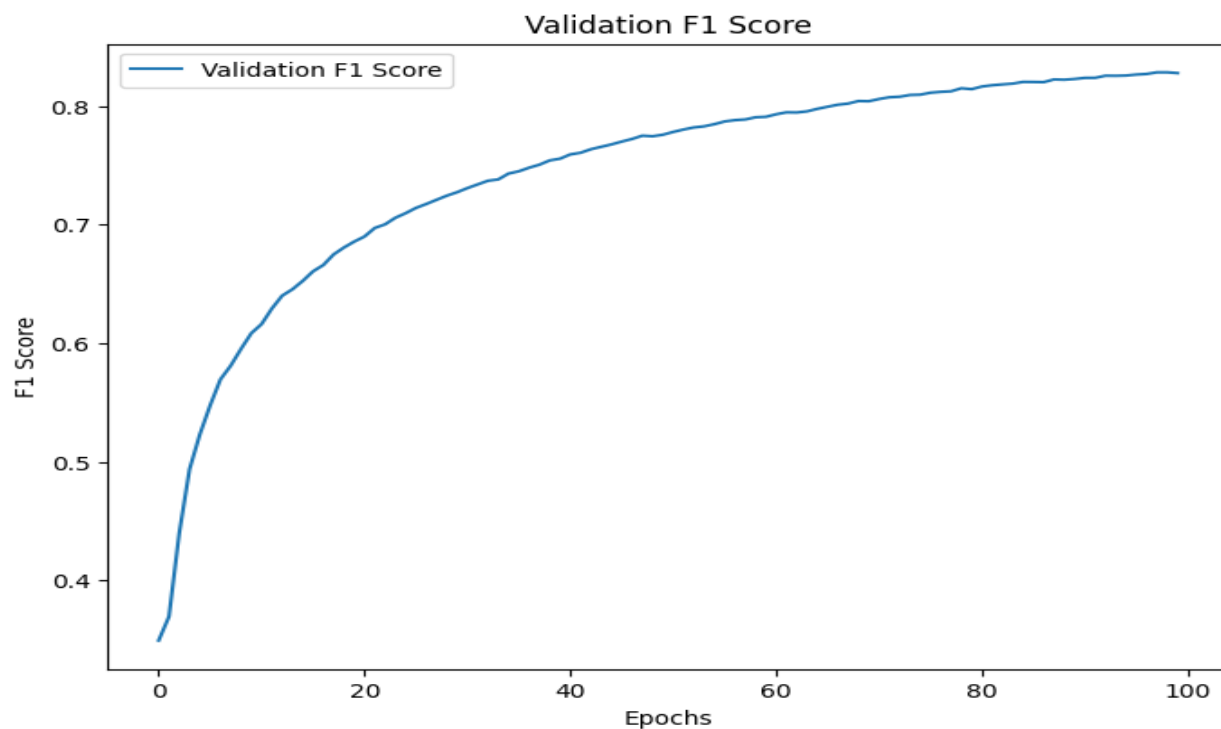
LR = 0.005



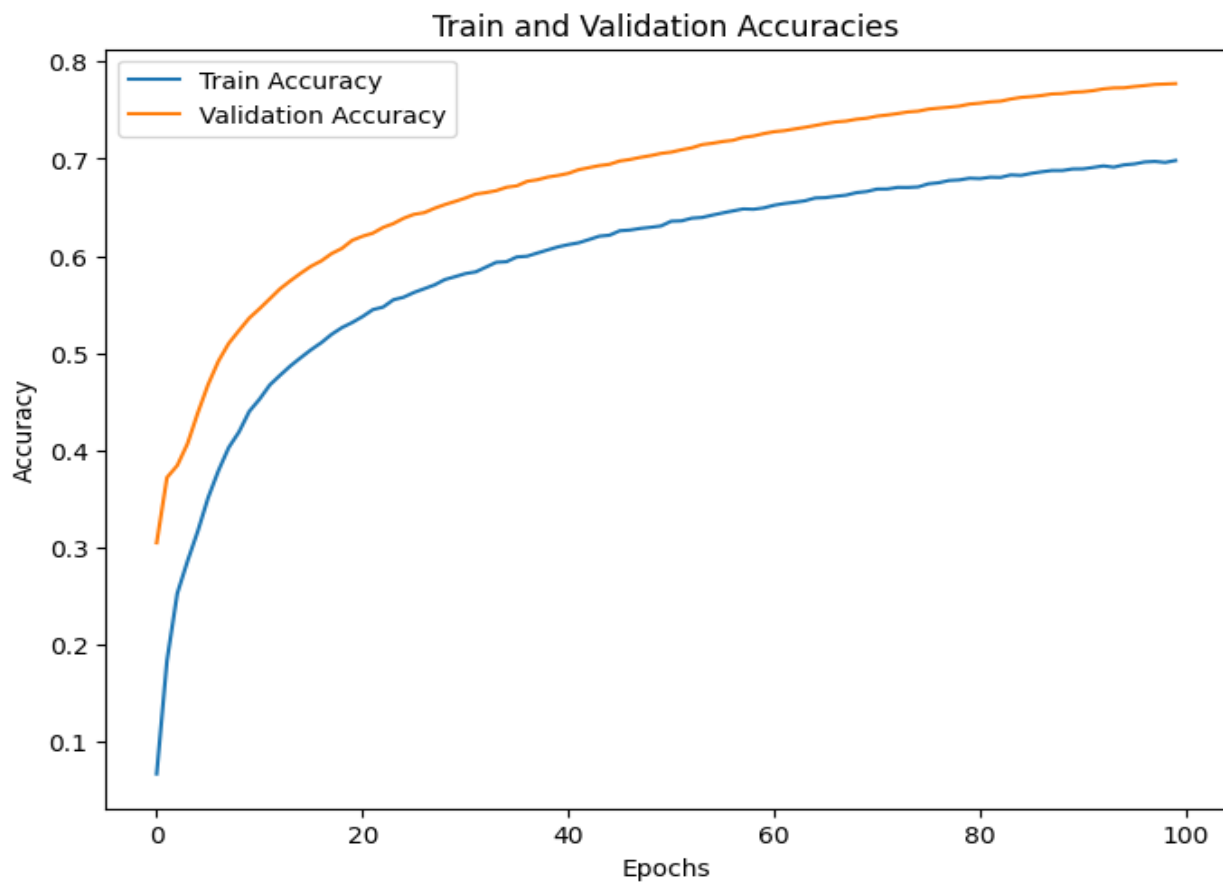
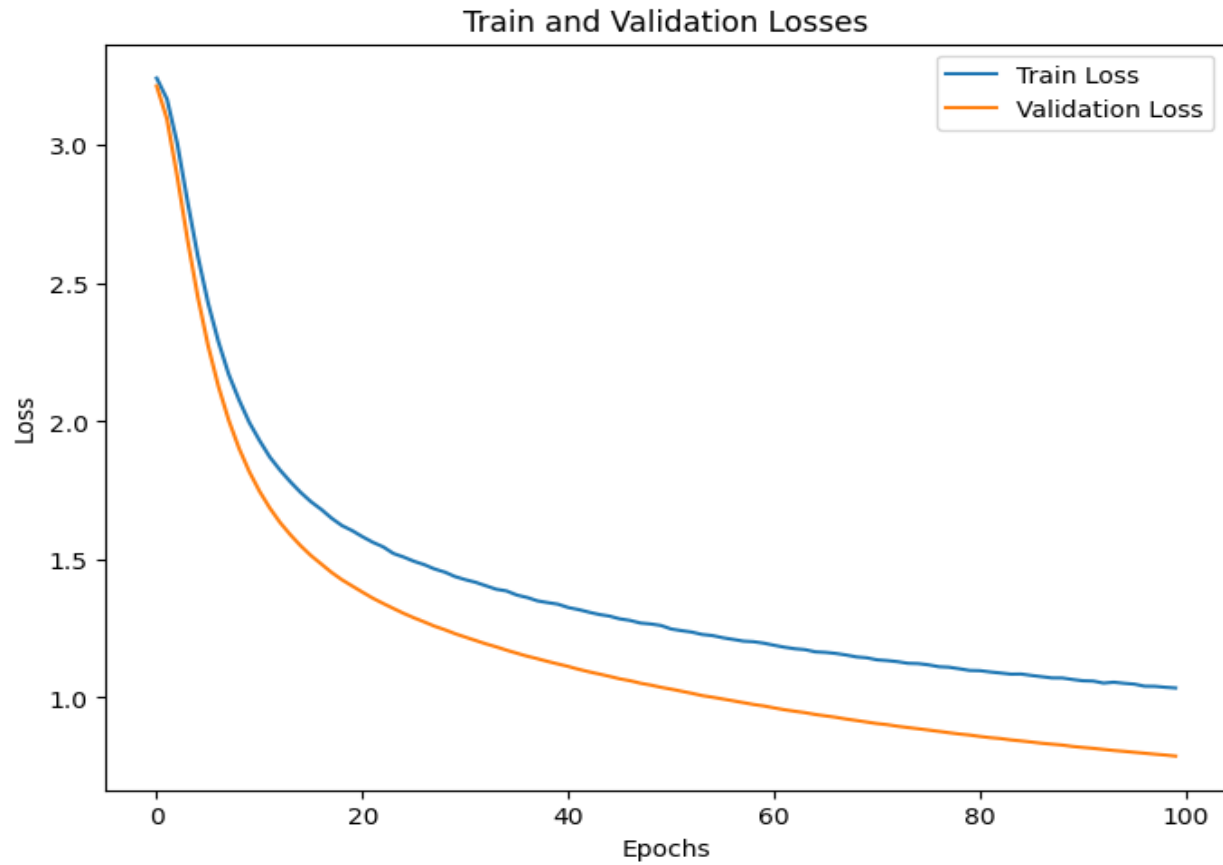


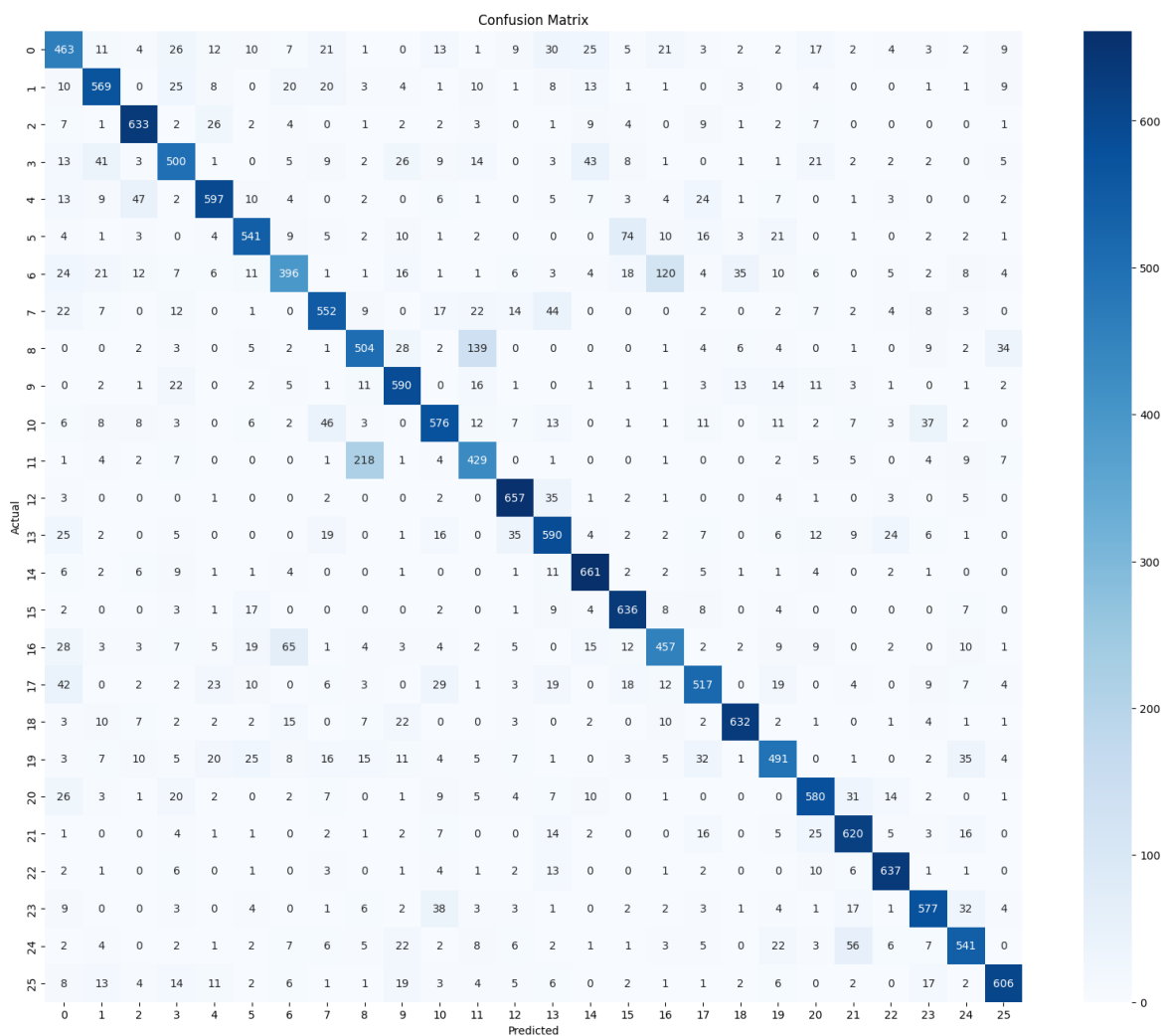
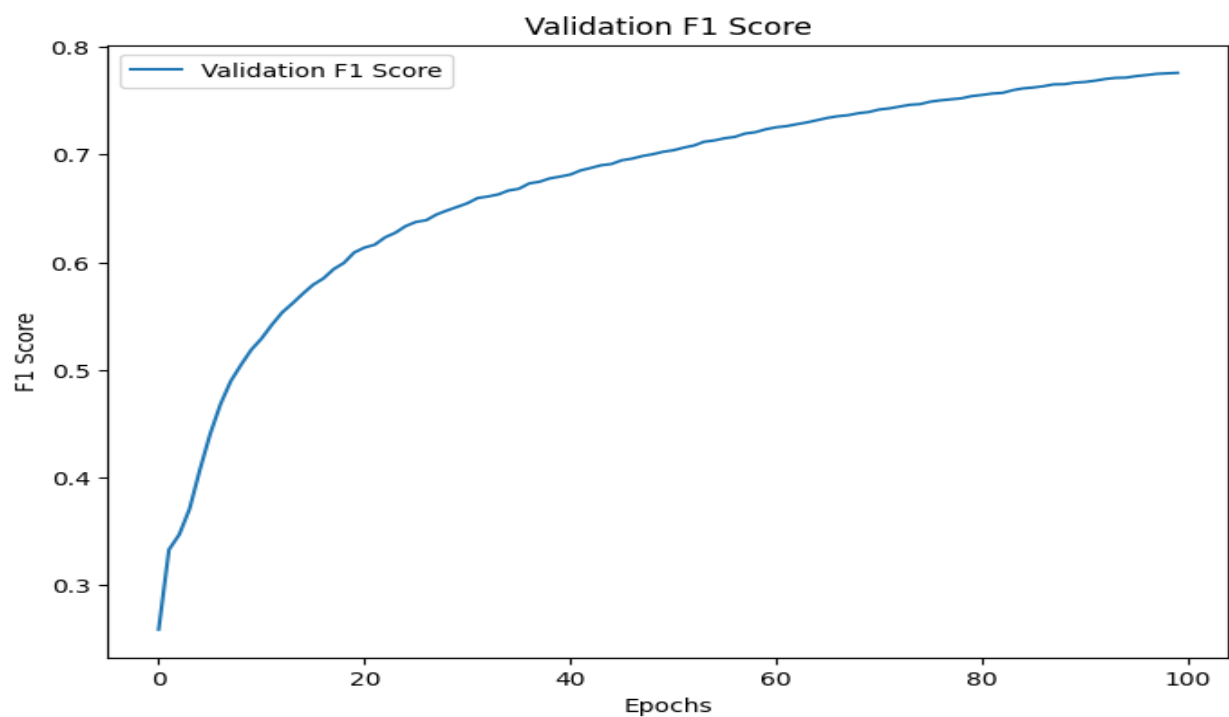
LR = 0.001





LR = 0.0005





Model 3:

```
L1=Dense(28*28, 256)
L2=ReLU()
L3=Dropout(0.25)
L4=Dense(256,26)

network = [
    L1,
    L2,
    L3,
    L4
]

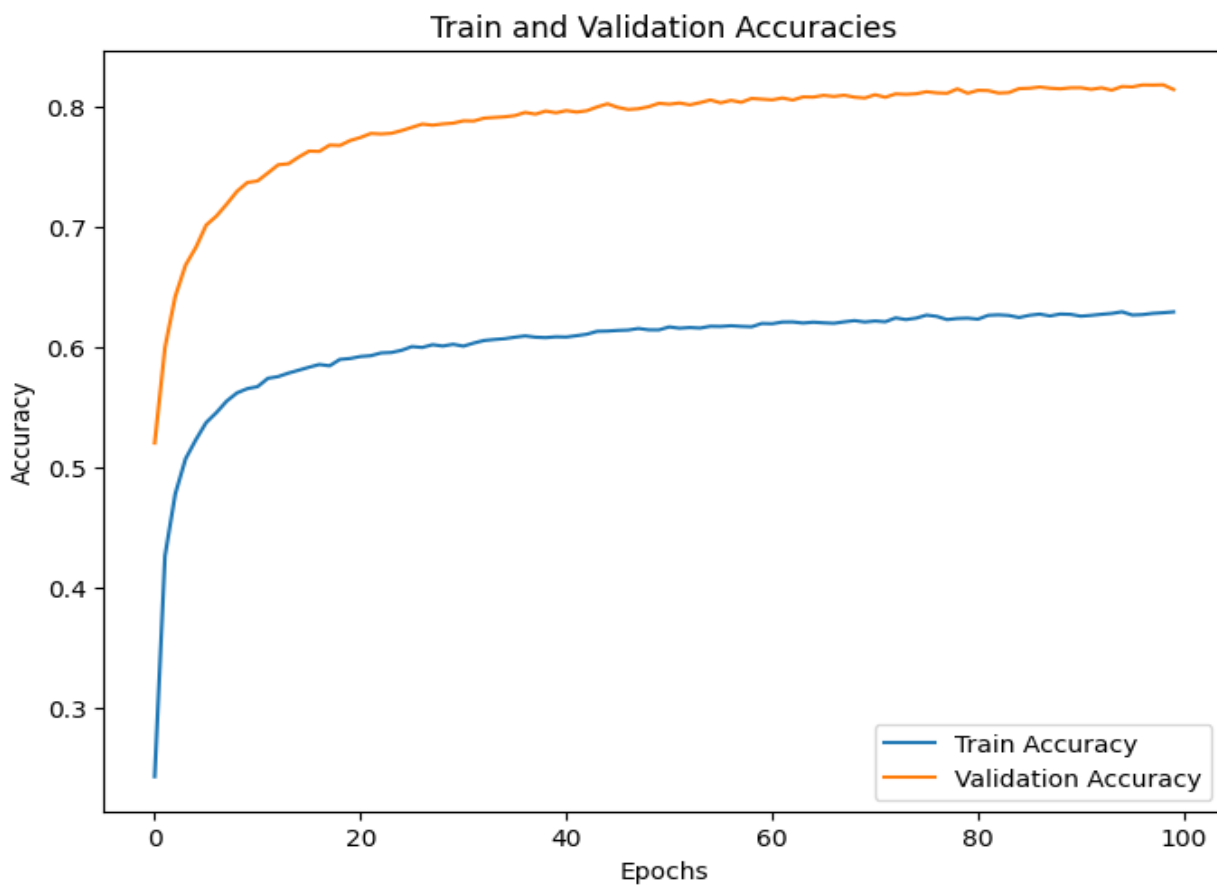
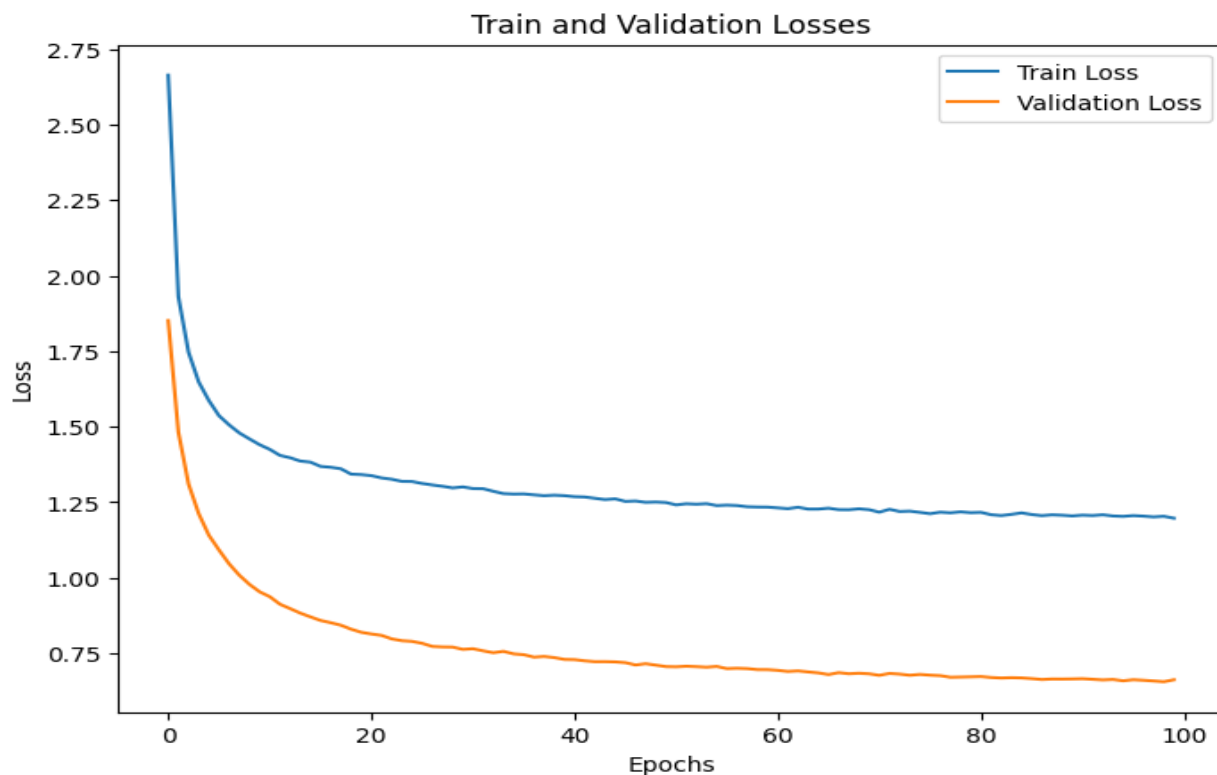
train(network, softmax_crossEntropy(), train_loader, val_loader, epochs=100, learning_rate=0.001)
```

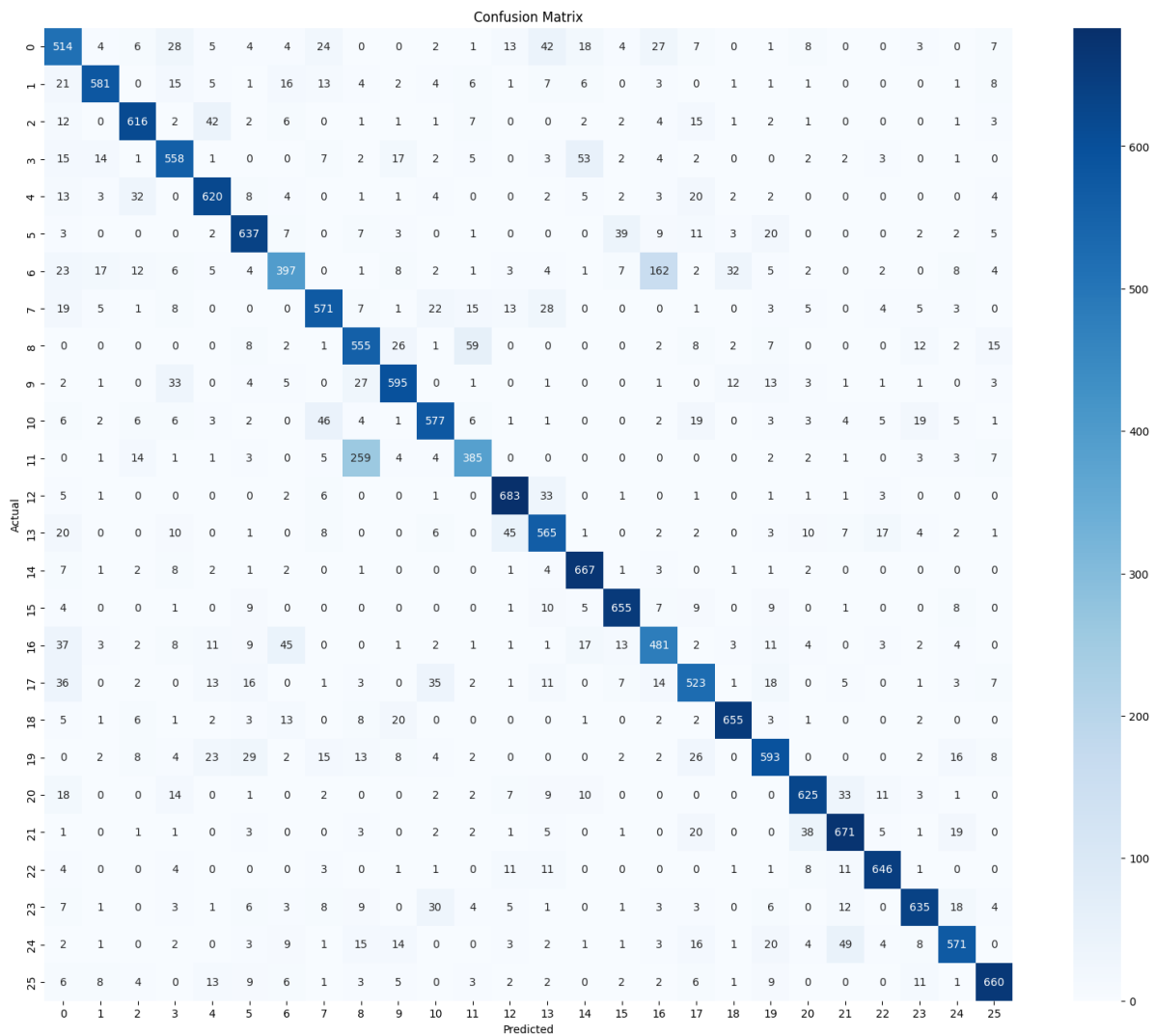
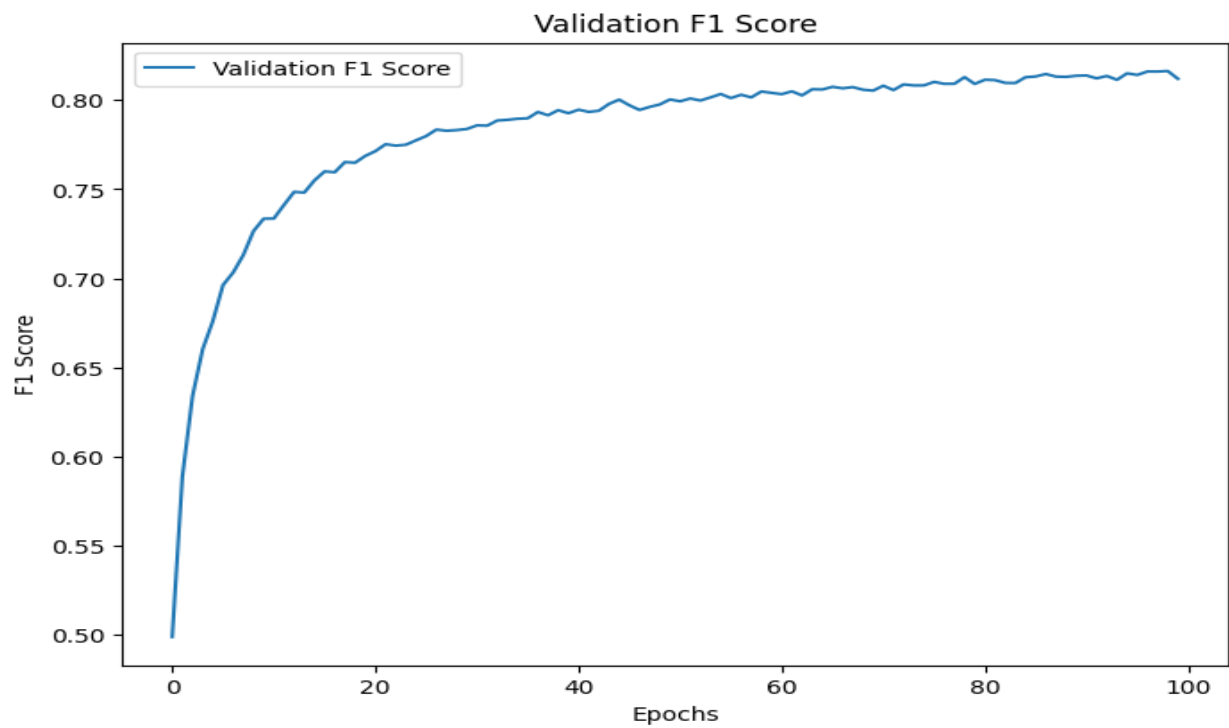
Metrics:

learning rate	loss		accuracy		macro_f1
	training	validation	training	validation	validation
0.01	1.203	0.655	0.628	0.818	0.816
0.005	1.057	0.609	0.678	0.826	0.825
0.001	1.219	0.804	0.636	0.773	0.773
0.0005	1.367	1.023	0.595	0.711	0.708

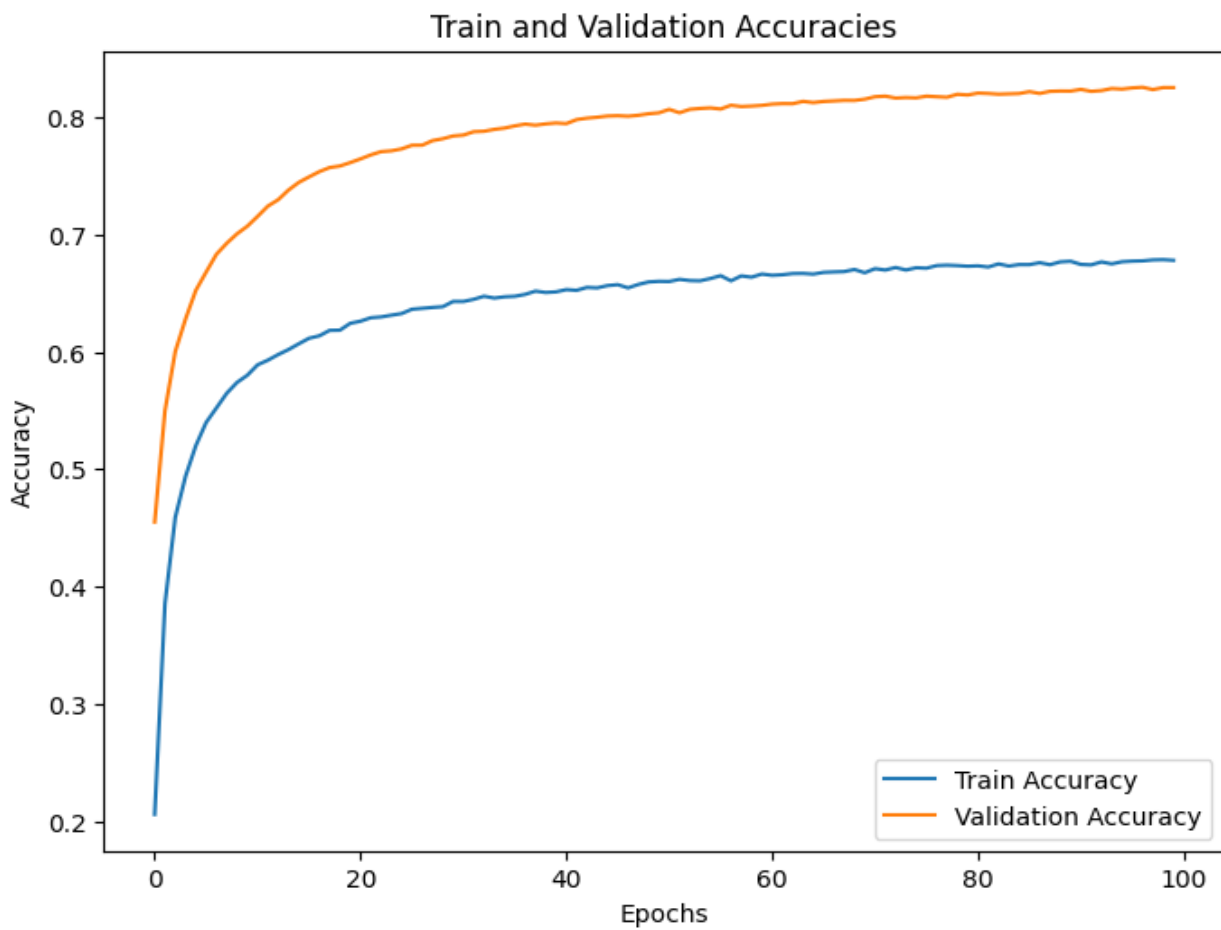
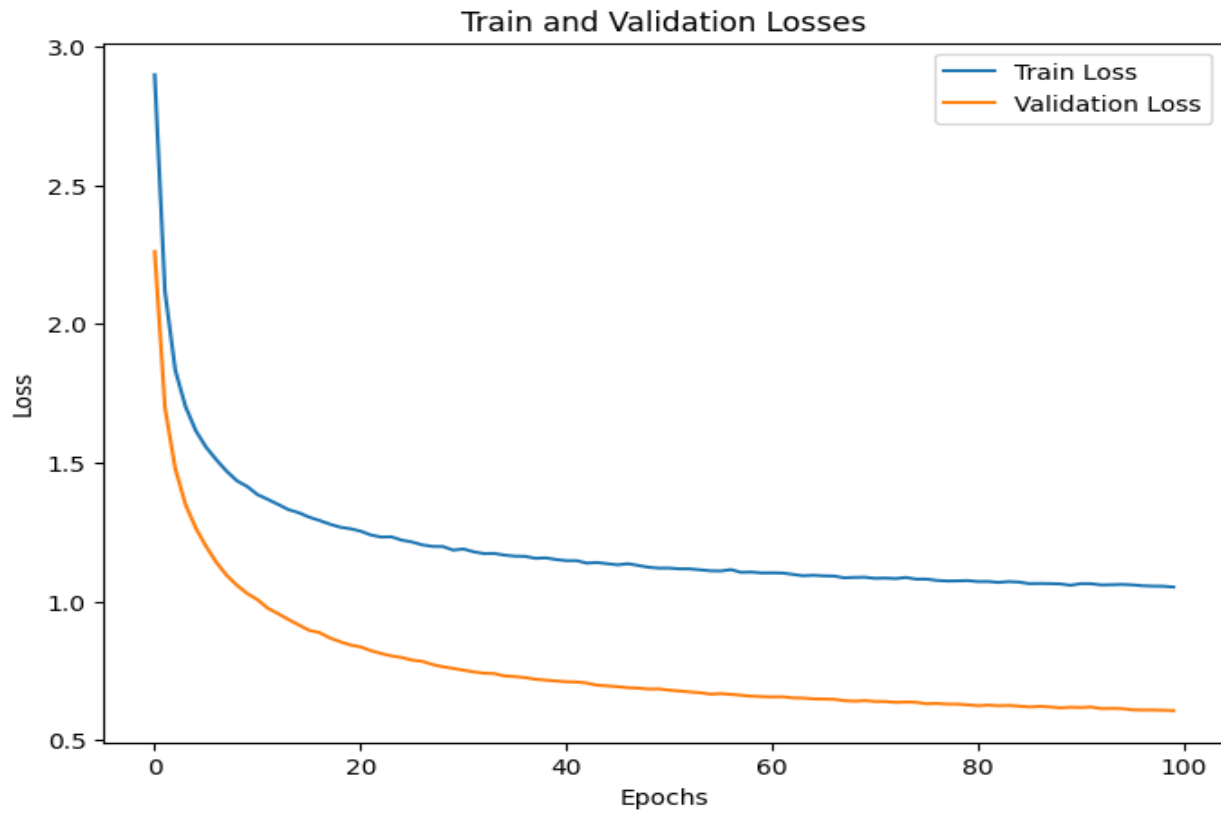
Graphs of Metrics:

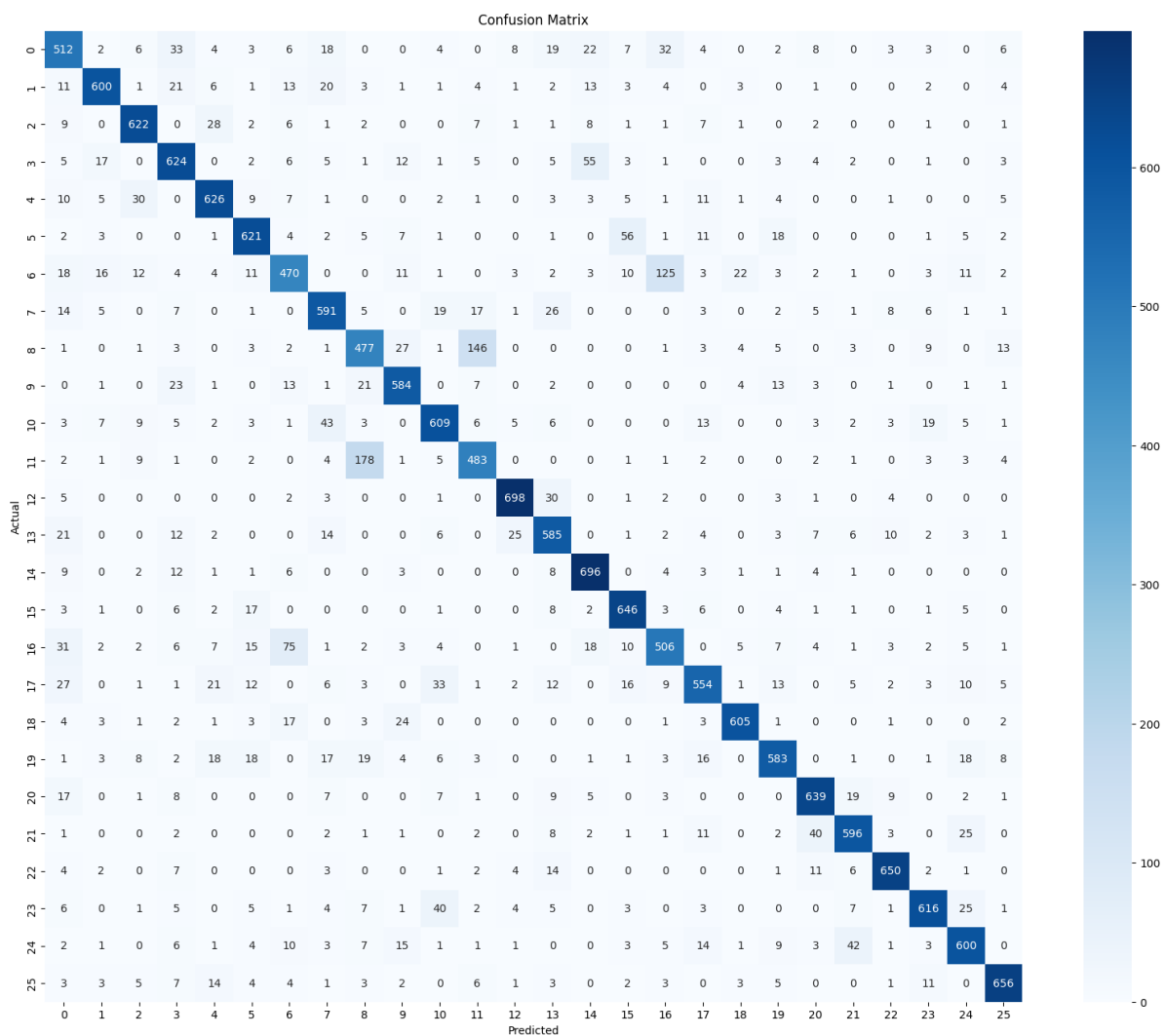
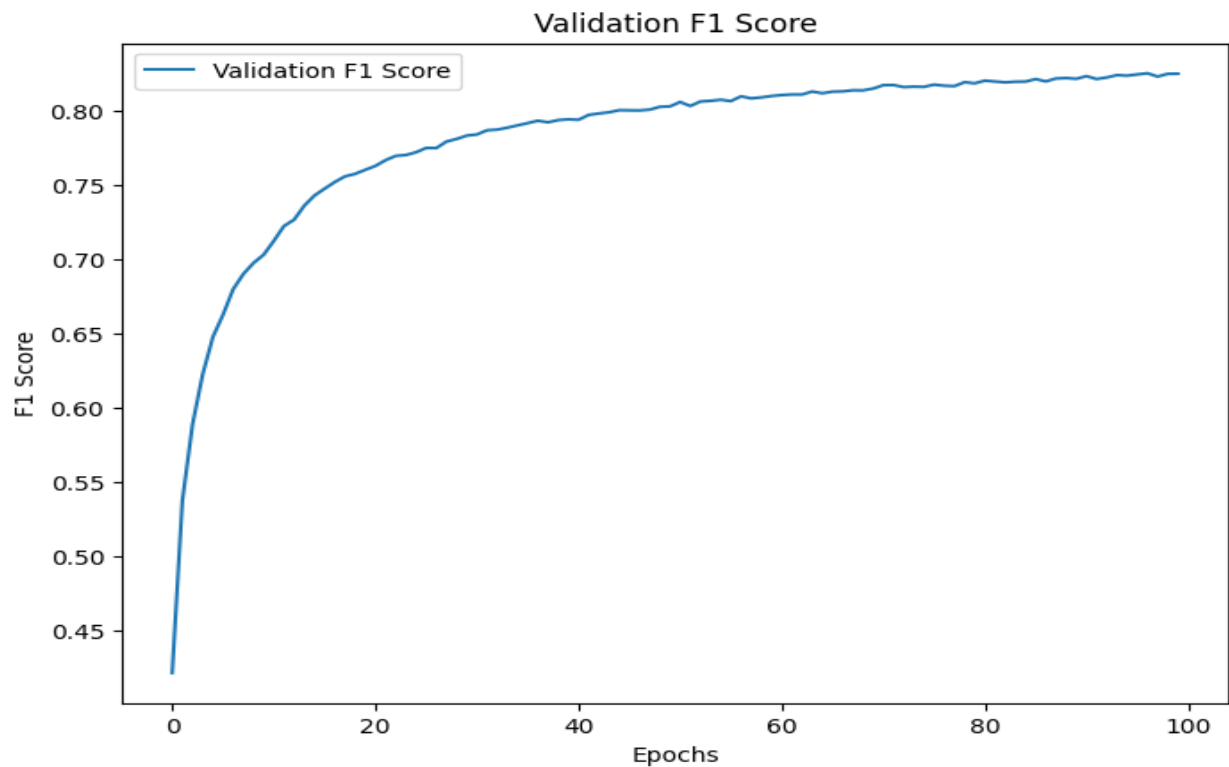
LR = 0.01



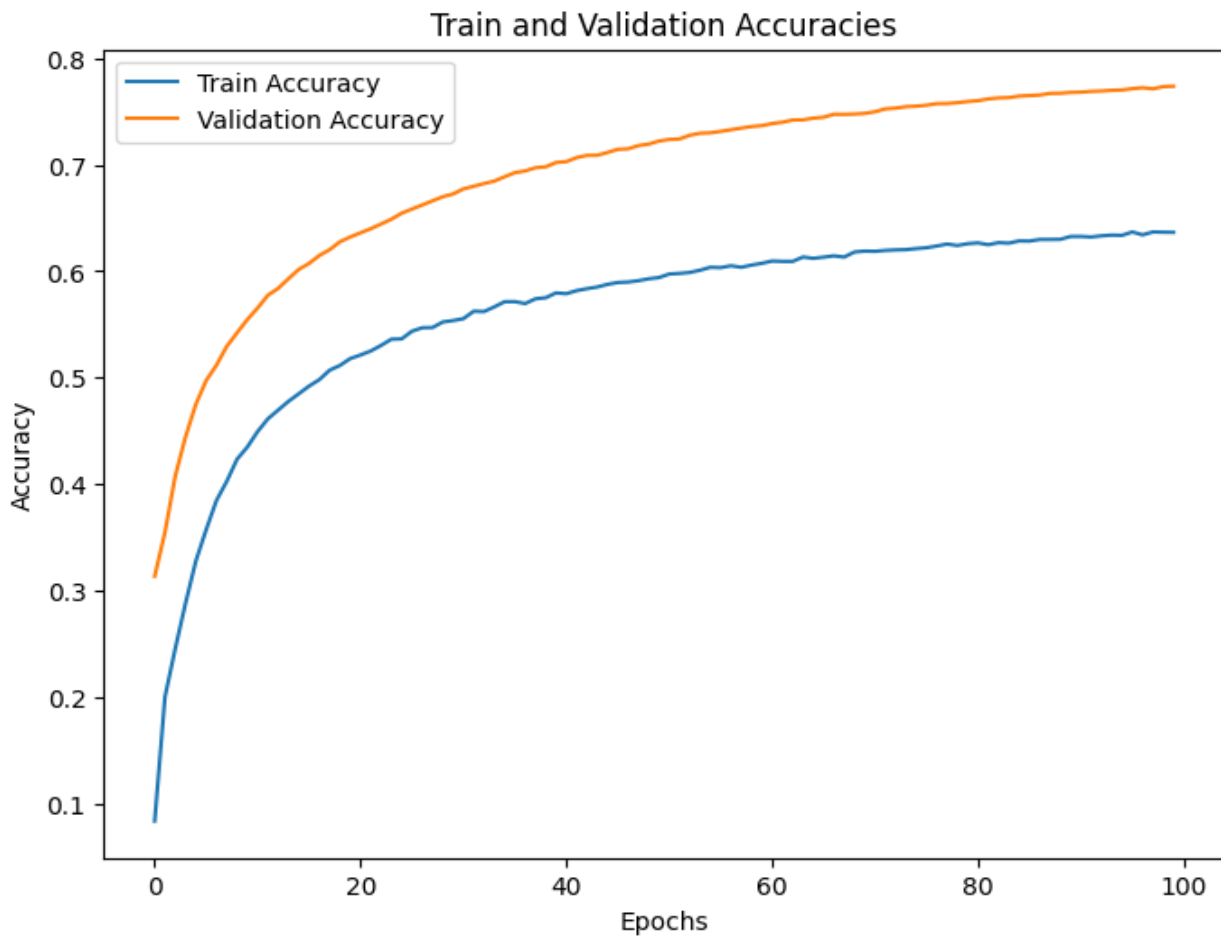
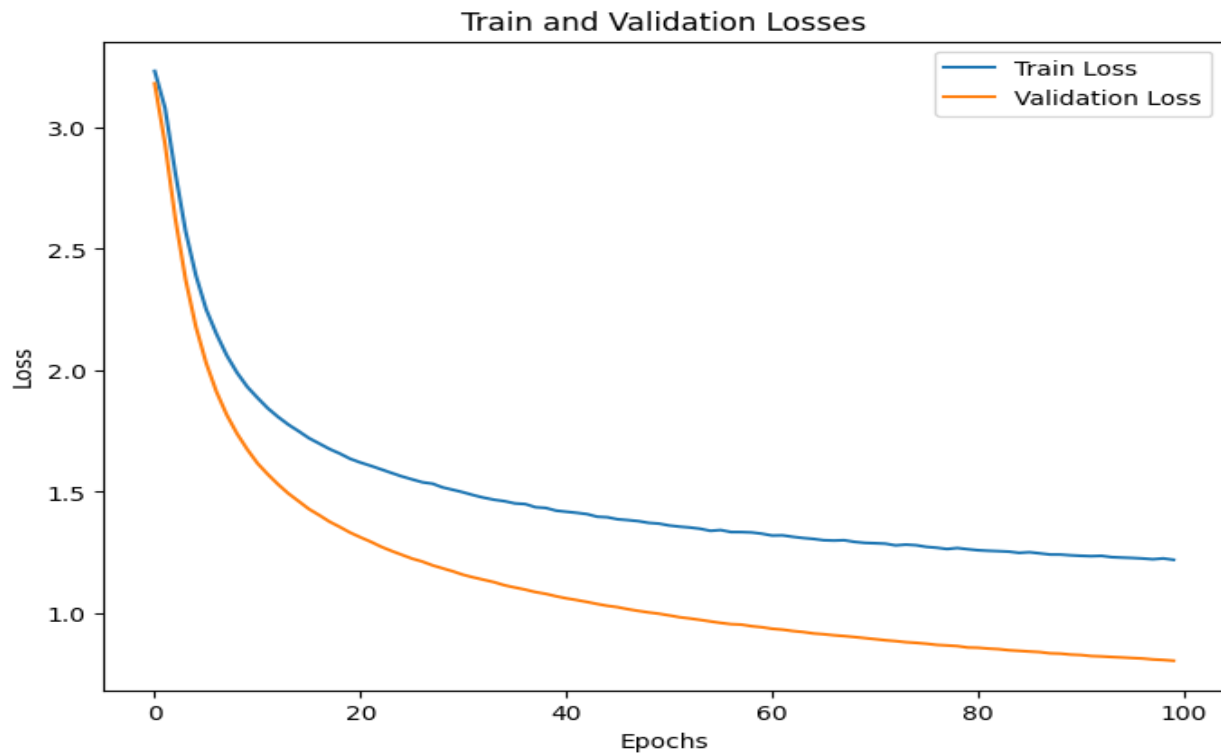


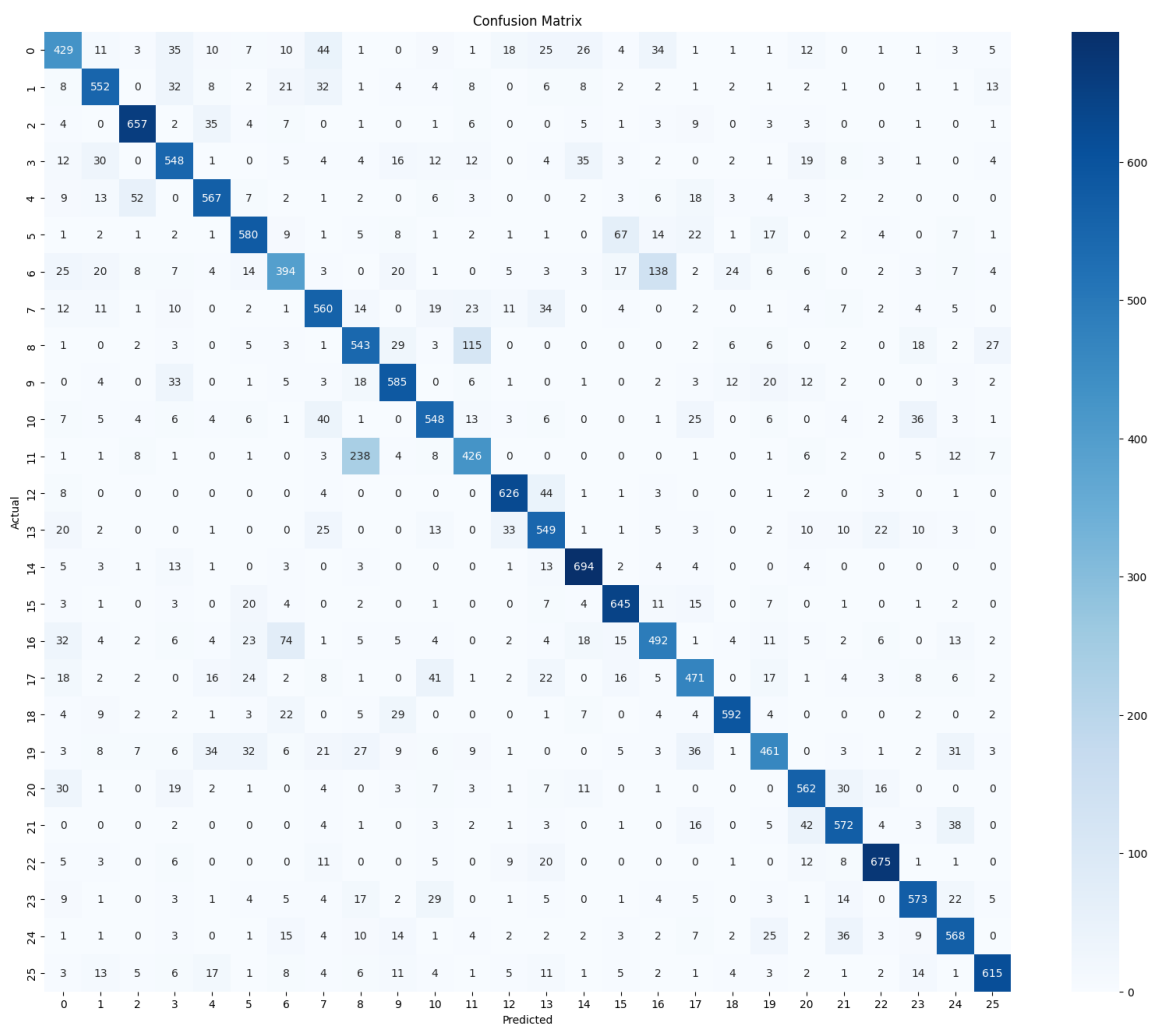
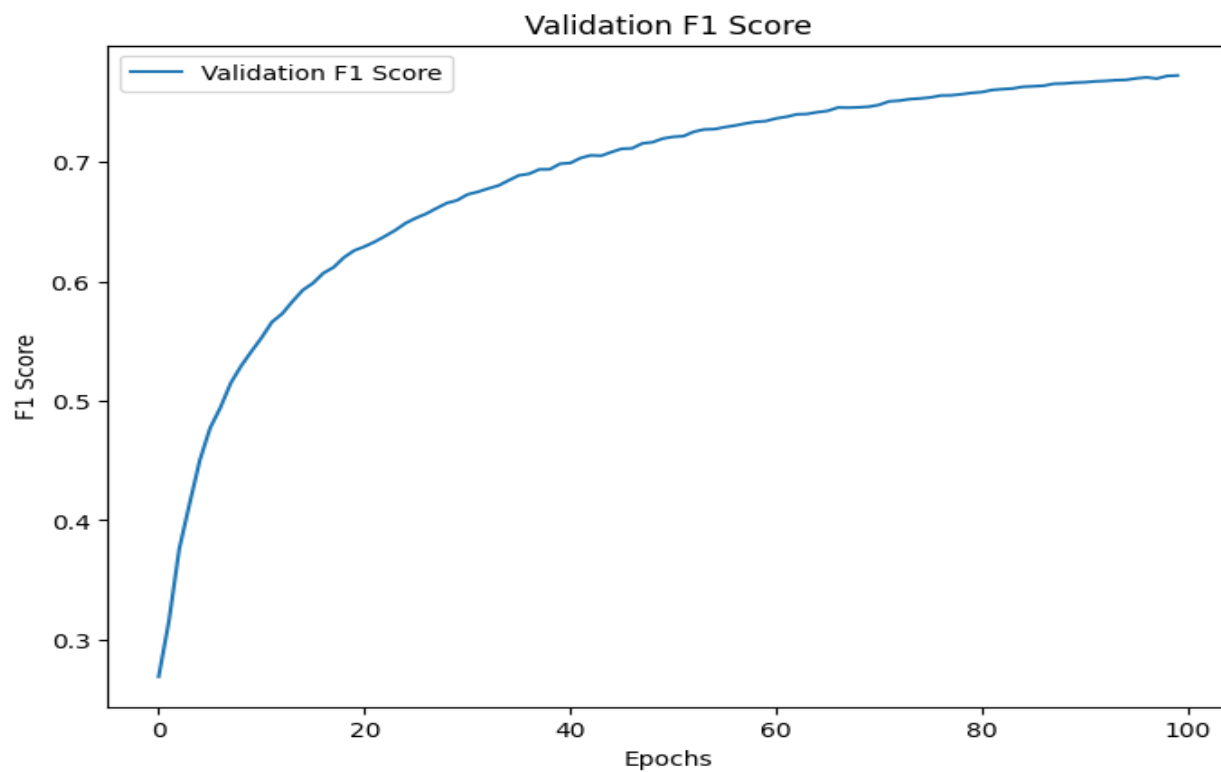
LR = 0.005



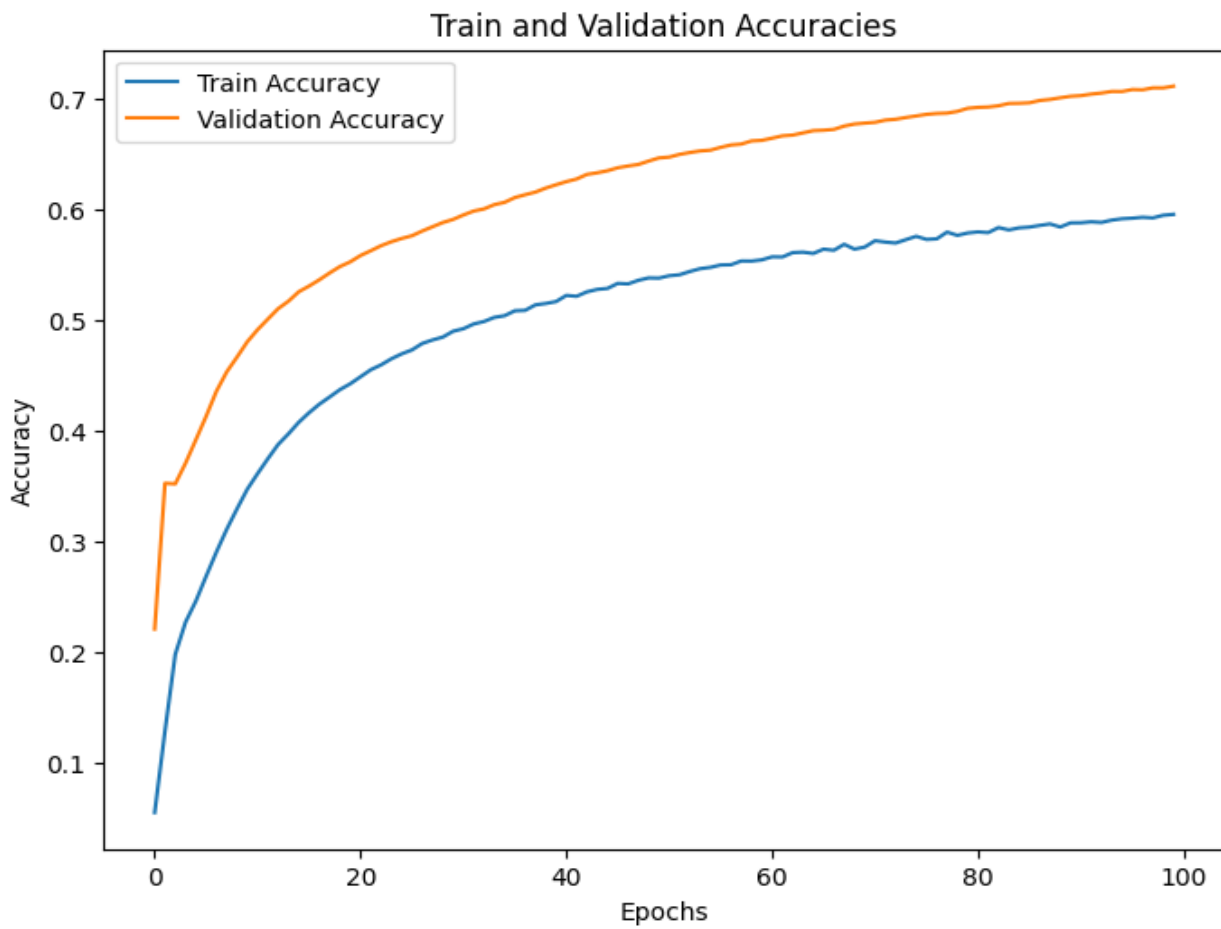
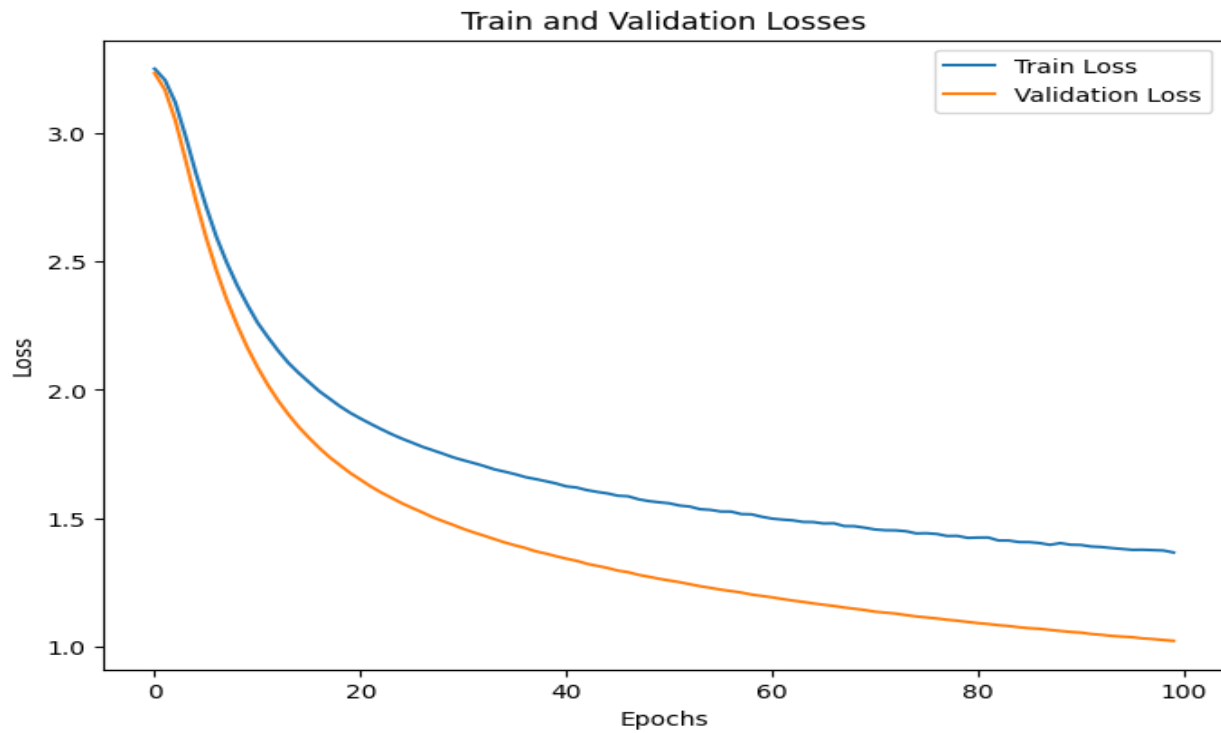


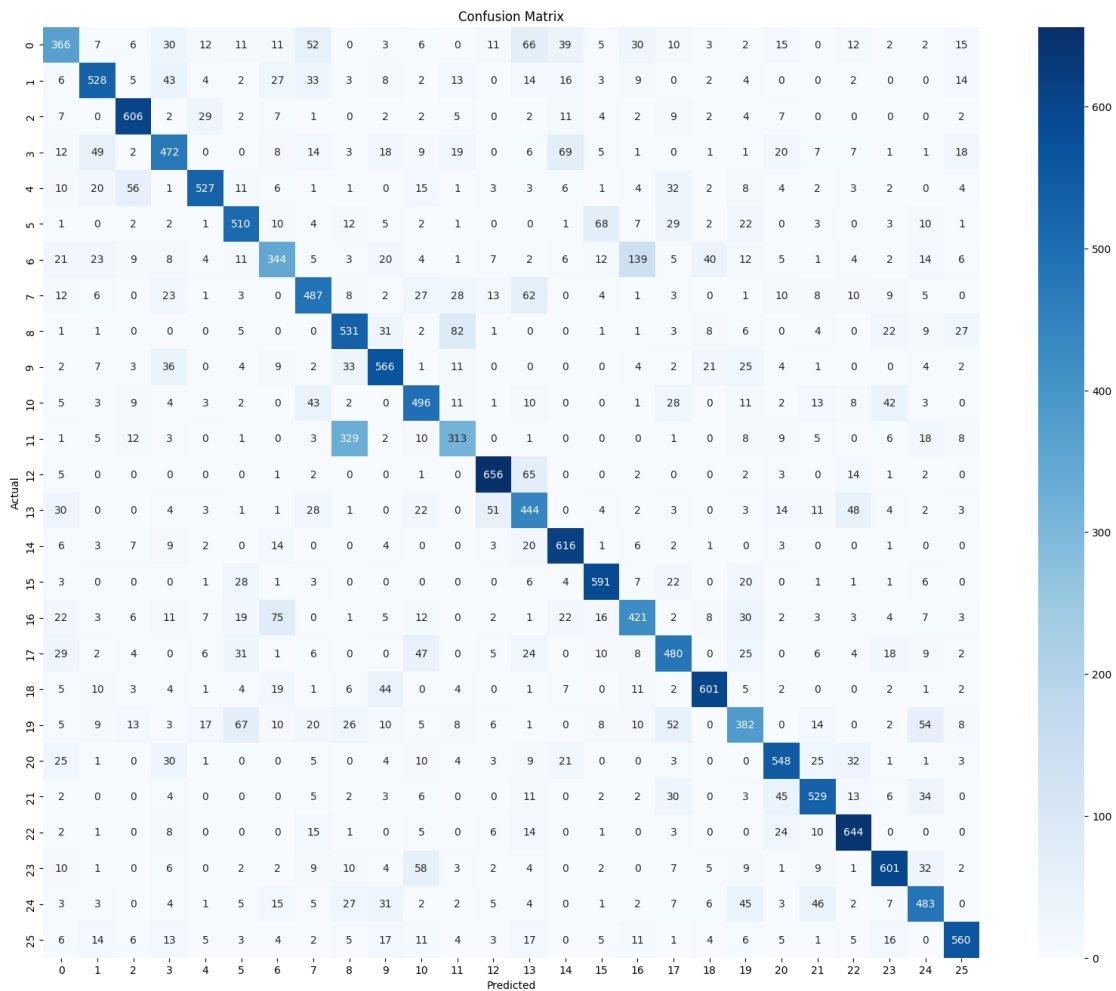
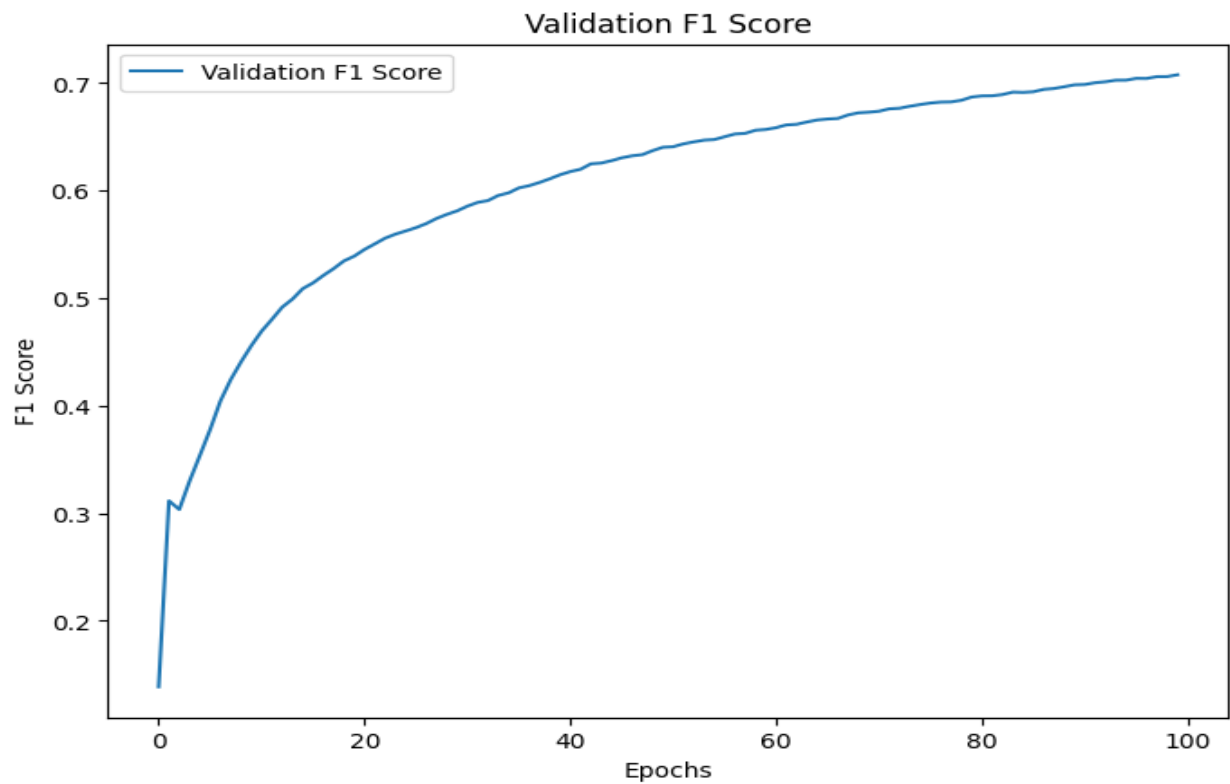
LR = 0.001





LR = 0.0005

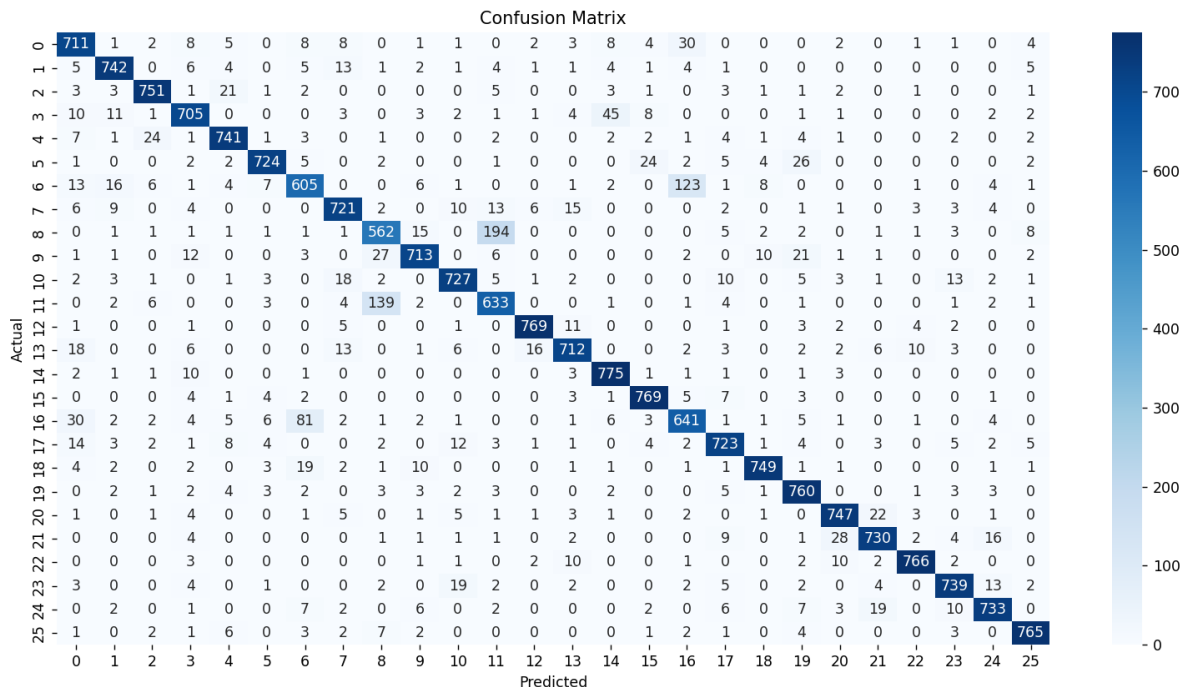




Independent Test Performance

➤ Model 1, LR = 0.01, epochs = 100

- Test Loss: **0.333**
- Test Accuracy: **0.8997**
- Test Macro_f1: **0.8996**
- Confusion Matrix:



Observations:

- While increasing the learning rate to 0.01, the accuracy graph plateaued earlier in the training process. This indicated that the model's performance didn't improve significantly after certain epochs. As a result, it seemed like the model might require more training epochs to further improve its accuracy. However, due to time constraints, I couldn't explore training the model for a longer duration to assess its performance better.
- Due to dropout usage, the training loss and accuracy weren't superior to those from the validation set.
- Increasing the number of nodes enhances accuracy, reduces loss, and improves validation F1 score.